

Highlights

Train once, deploy across hospitals? A domain-complete evaluation and an evidence-graded roadmap for cross-site generalization in computational pathology

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- Models trained once can work at unseen sites, but no recipe wins everywhere
- Judge a pathology model by its worst hospital or scanner, not by its average
- Default recipe: fine-tune a pathology foundation model with color jitter
- A site acceptance test checks each new hospital with a few local labels
- At standard scale it needs about half the labels, with $\leq 2.1\%$ false acceptance

Train once, deploy across hospitals? A domain-complete evaluation and an evidence-graded roadmap for cross-site generalization in computational pathology

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ABSTRACT

Practitioners want pathology models that work at new hospitals and scanners without retraining, yet the methods proposed for this are usually compared on one held-out site with one training run each. We test the train-once strategy directly and propose a statistical test for accepting each new site. Every domain of four public cohorts that cross the task (tumor versus mitotic figure) with the type of shift (scanners versus institutions) is held out in turn, and up to 21 methods, including pathology foundation models, are run with several seeds at two model scales (4,778 runs). Without retraining, the best standard-scale method of each cohort reached a worst-site area under the ROC curve (AUROC) of 0.88–0.99, against 0.53–0.90 for empirical risk minimization (ERM). No method was best everywhere, none beat ERM after correction for multiple comparisons, and the held-out site explained most of the variance of a run. Fine-tuned foundation models with color augmentation were the most reliable choice on average. Because development results cannot guarantee performance at a new site, we propose an empirical-Bayes site acceptance test. It combines the spread of performance across development sites with a small labeled sample from the new site and accepts the site if its AUROC is likely to reach a target. Validated on every held-out site at standard scale, it kept false acceptance at or below 2.1% and needed about half as many labels as local labels alone, whereas the development results alone accepted 8.6% of the sites below a target AUROC of 0.80. We condense the findings into a six-step, evidence-graded roadmap for building, adapting, validating and accepting models across sites.

1. Introduction

Deep networks for histopathology often lose accuracy when they are applied to slides from a laboratory, scanner or patient population they were not trained on. Staining protocols, scanners and tissue handling differ between sites, and networks pick these differences up: the submitting site of a TCGA slide can be predicted from the image, and this signal biases downstream predictions (Howard et al., 2021). Color differences between laboratories degrade classifiers on unseen laboratories (Tellez et al., 2019), and mitotic-figure detectors deteriorate on new scanners and tumor types (Aubreville et al., 2023a, 2024). The problem is not specific to pathology; most externally validated radiology models also perform worse on external data (Yu et al., 2022).

For practitioners, the goal is a model that is trained once and then works at every hospital where it is used, without collecting labels or retraining at each new site. Many remedies promise this: stain normalization and color augmentation, domain-generalization (DG) objectives, self-supervised pre-training, test-time adaptation and, most recently, pathology foundation models. To choose between them, practitioners need evidence that can tell a real difference from noise. That is harder than it looks. Benchmarks with a single official test site, such as Camelyon17-WILDS, evaluate every method on one domain, and ERM accuracy on that domain has a standard deviation of 6.4 points across ten seeds (Koh et al., 2021). Across DG benchmarks, ERM is hard to beat once every method is tuned and evaluated under the same protocol (Gulrajani and Lopez-Paz, 2021). Rankings in biomedical image analysis challenges are sensitive to the test data and the ranking scheme (Maier-Hein et al., 2018). A reported gain can thus reflect the chosen test site, the seed or the tuning budget as much as the method.

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This study began with such a gain. In an earlier version of this work, a single Camelyon17 test hospital suggested that self-supervised pre-training and sharpness-aware minimization were synergistic. Each method alone was worse than ERM, and together they were better. Repeating the runs and holding out every hospital in turn reversed the sign of this interaction at most hospitals (Section 6.4). The same earlier version argued that the type of shift, scanner-only or institutional, decides which method works best. That argument rested on two cohorts that differed in task as well as in shift type, and an audit of our code found defects in several method implementations. We therefore rebuilt the study around six questions:

1. How much of the variance of a held-out score comes from the choice of held-out domain, and how much from run-to-run noise?
2. With the few domains that pathology cohorts provide, can realistic differences between methods be detected at all?
3. How stable are method rankings within a cohort?
4. Do rankings carry over between cohorts that share the task, or between cohorts that share the type of shift?
5. Which choices give a model the best chance of working at an unseen site without retraining?
6. How can a practitioner decide, with few local labels, whether a trained model is good enough at a particular new site?

To answer them, we use a *domain-complete, seed-replicated* protocol. Every domain of a cohort is the test domain of one leave-one-domain-out fold, every method is run at several seeds under one fixed training budget, and hyperparameters and checkpoints are selected without test data. We analyze the results with a random-effects decomposition into between-domain and between-run variance with interval estimates, paired contrasts with a fold-dependence sensitivity analysis, a power calculation based on the variance of the paired difference, and bootstrap intervals for ranks and for rank agreement between cohorts. We apply the protocol to four public cohorts arranged in a 2×2 design. Two tumor-classification cohorts and two mitotic-figure cohorts each pair an acquisition-only shift (scanners within one laboratory) with an institutional shift (hospitals or tumor types). We evaluate up to 21 arms at two model scales, a compact CNN and ImageNet ResNet-50, together with frozen and fine-tuned pathology foundation models, in 4,778 training runs. No model is ever trained on labeled data from its test site, and apart from the test-time adaptation arms, which use unlabeled test images, every result measures how a model trained once performs, unchanged, at a site it has never seen.

Our contributions are:

- an empirical-Bayes *site acceptance test* that decides whether a trained model reaches a target AUROC at a new site. It combines the between-site spread of the development results with a small labeled sample from the new site, plans the number of labels in advance, and is validated on every held-out site of the study (Sections 3.7 and 5.7);
- a six-step, evidence-graded roadmap for practitioners who need models that work across hospitals and scanners without retraining, covering the target definition, data collection, training recipe, label-free adaptation at a new site, validation and site acceptance (Section 6.2);
- an evaluation protocol and a set of diagnostics for distribution shift in multi-domain pathology cohorts, with the code, the fold definitions and every per-run record released so that each table and figure can be regenerated;
- a 2×2 task-by-shift study at two model scales, with domain-generalization objectives, stain and color methods, self-supervision, test-time adaptation, pathology foundation models and two controls for self-supervision, all tuned per fold;
- an external reference point for the variance decomposition, computed from DomainBed’s published per-environment results;
- two negative results reported in full: a single-hospital synergy claim that does not survive the protocol, and a single-institution cohort that cannot support domain claims.

The main findings are these. Without retraining, the best standard-scale method of each cohort reached a worst-site AUROC of 0.884 to 0.986, against 0.534 to 0.895 for ERM, but no method was best everywhere. For most arms, the

held-out domain accounts for most of the variance of a single run (median ICC 0.72–0.95), and the worst domain differs between methods. After correction for multiple comparisons, no standard-tier method beats ERM on any cohort. Even a mean gain of 0.145 AUROC that holds at every domain is not significant with five domains, and detecting a gain of 0.05 on the tumor cohorts would need between 7 and 53 domains. Under the primary selection rule, none of the six DG objectives ranks in the top five on any cohort. At the standard tier, method rankings agree more between cohorts that share a task than between cohorts that share a shift type. With one cohort per cell this is a description, not a proof, and the compact tier does not clearly show it. Fine-tuned pathology foundation models with color augmentation were the most reliable choice on average, and re-estimating batch-normalization statistics at the new site, which needs no labels and no weight updates, helped mainly under severe scanner shift. Implementation choices such as per-tile versus per-slide stain estimation, and the learning rate of a pretrained encoder, change results as much as the choice of method. Finally, the development results alone would have accepted 8.6% of the sites that fell below an AUROC of 0.80. At standard scale, the site acceptance test kept false acceptance at or below 2.1% and, for targets of 0.80 and 0.85, reached the same power as local labels alone with about half as many labels.

Section 2 reviews related work. Section 3 defines the protocol, the statistics and the site acceptance test, and Section 4 the cohorts, models and arms. Section 5 reports the results. Section 6 discusses them, presents the roadmap (Section 6.2) and the two case studies, and lists the limitations.

2. Related work

2.1. Site and center effects

Performance loss at institutions not seen in training is documented across medical imaging. Zech et al. (2018) trained pneumonia classifiers on chest radiographs from three hospital systems. Internal performance exceeded external performance in three of five comparisons, and for two of the systems the networks identified the system a radiograph came from with more than 99.9% accuracy. In a systematic review of 86 externally validated radiology algorithms, 81% performed worse on external data and 24% lost at least 0.10 on the unit scale (Yu et al., 2022). In histopathology, Howard et al. (2021) showed that the submitting site of TCGA slides can be identified by deep networks even after common color normalization and augmentation. These site signatures biased the accuracy of survival, mutation and stage prediction. The authors proposed splits in which no site contributes to both training and validation. Stacke et al. (2019) proposed a model-specific distance between domains, computed in the learned representation. Stacke et al. (2021) developed it into the *representation shift* and showed that it correlates highly with the drop in performance across many types of domain shift. These studies show that site effects exist and can be detected, but not how much of the spread in reported performance they account for relative to the training method. Our variance decomposition estimates this.

Mitotic-figure detection has been studied directly under domain shift. The MIDOG 2021 challenge trained on 200 breast cancer cases from four scanners (one of them without labels) and tested on 100 cases from four scanners, two of which were not seen in training. The winning method reached an F1 score of 0.748 (Aubreville et al., 2023a). MIDOG 2022 extended the shift across tumor types, laboratories and species (as its title states) and evaluated nine methods on ten independent domains. Domain characteristics absent from the training data (a new species, a new cell morphology and a new scanner) led to small but significant drops in performance (Aubreville et al., 2024). The method that won both editions combined a two-stage detector with domain-generalization measures (Jahanifar et al., 2024). MIDOG++ (Aubreville et al., 2023b) released 503 images of seven tumor types and found that leave-one-domain-out training generalized considerably better than training on a single domain. The multi-scanner canine cutaneous squamous cell carcinoma dataset (Wilm et al., 2023) digitized the same slides with five scanners, which isolates the acquisition shift. We use these cohorts, but our question is not the challenges' question of which detector is best. We ask how much of a comparison between methods is decided by the domains themselves.

2.2. Domain generalization benchmarks and protocols

Leave-one-domain-out evaluation with repeated runs is established practice in domain generalization (DG); Zhou et al. (2023) survey the field. DomainBed (Gulrajani and Lopez-Paz, 2021) evaluated fourteen algorithms on seven multi-domain datasets. It trained on the remaining domains for every choice of held-out test domain, and it repeated the whole study three times, redrawing hyperparameters, weight initializations and data splits. The authors argued that a DG algorithm without a model-selection rule is incomplete. They compared three rules: validation data pooled from the training domains, leave-one-domain-out cross-validation over the training domains, and an oracle validation set

drawn from the test domain. With training-domain or oracle selection, no algorithm improved on the average accuracy of empirical risk minimization (ERM) by more than one point, and the authors found no improvement over ERM statistically significant. Under leave-one-domain-out selection, however, CORAL led ERM by 1.8 points (Gulrajani and Lopez-Paz, 2021, Table 3). Training-domain validation also outperformed leave-one-domain-out cross-validation across multiple datasets and algorithms.

WILDS (Koh et al., 2021) made Camelyon17 (Bandi et al., 2019) a DG benchmark with three training hospitals, one out-of-distribution (OOD) validation hospital and one test hospital. The test hospital was chosen because its patches were the most visually distinctive. ERM test accuracy had a standard deviation of 6.4 points over ten seeds, and CORAL, IRM and GroupDRO performed comparably to or worse than ERM. WILDS therefore asks for ten seeds per Camelyon17 submission and warns that so few test domains invite overfitting to them. WILDS 2.0 (Sagawa et al., 2022) added unlabeled patches from the training, validation and test hospitals. On Camelyon17, ERM reached 70.8% OOD accuracy without data augmentation and 82.0% with it. SwAV pre-training on unlabeled target-hospital patches reached 91.4% (standard deviation 2.0) and Noisy Student reached 86.7%. CORAL, DANN, Pseudo-Label and FixMatch all fell below augmented ERM (Sagawa et al., 2022, Table 2). These figures come from one held-out hospital. The best arm also sees unlabeled images from that hospital, so it is unsupervised adaptation rather than DG. Wiles et al. (2022) distinguished spurious correlation, low-data drift and unseen data shift, and trained more than 85,000 models covering 19 methods on six datasets, Camelyon17 among them. Pre-training and augmentation often helped substantially, but the best methods differed across datasets and shifts. On Camelyon17, color jitter hurt performance, and pre-training was not the best choice under unseen data shift.

We adopt leave-one-domain-out evaluation and training-domain model selection from this literature and claim neither as new. Our contribution is diagnostic. We decompose performance into between-domain and between-run variance with interval estimates, and we analyze how stable method rankings are across held-out domains and cohorts. We use a factorial cohort design that crosses task (tumor versus mitotic figure) with shift type (institutional versus acquisition-only). Ranking agreement can then be compared between cohorts that share a task and cohorts that share a shift type.

2.3. Stain normalization and augmentation

Differences in staining between laboratories degrade networks applied to slides from an unseen laboratory (Tellez et al., 2019). Reinhard et al. (2001) transferred color by matching per-channel means and standard deviations in the decorrelated $l\alpha\beta$ space. Color deconvolution (Ruifrok and Johnston, 2001) separates up to three stains in RGB images using stain-specific absorption, and it is the basis of HED-space augmentation (Tellez et al., 2019). Macenko et al. (2009) estimate the stain vectors of each slide from its optical densities. Pixels are projected onto the plane of the two leading singular vectors, the vectors are taken at robust angular extremes, and per-stain intensities are rescaled to a common pseudo-maximum. Vahadane et al. (2016) factorize images into sparse, non-negative stain density maps and recombine them with a target image's stain color basis. StainGAN (Shaban et al., 2019) learns the mapping with a CycleGAN-style model, so no single reference slide has to be chosen. Instead of normalizing, Tellez et al. (2018) perturbed the hematoxylin and eosin channels directly and, together with ensembling, obtained a stain-invariant mitosis detector. Across four tasks and nine laboratories, Tellez et al. (2019) found HSV or HED color augmentation essential. The top-ranked configuration used no normalization, although normalization made classifiers less sensitive to the choice of augmentation. Normalization also leaves site information in place (Howard et al., 2021). We therefore evaluate per-slide Macenko normalization, a per-tile variant, H&E jitter and their combination as separate arms.

2.4. Self-supervision and pathology foundation models

Ciga et al. (2022) pre-trained SimCLR on 57 unlabeled histopathology datasets. Linear classifiers on these features beat ImageNet features by more than 28% in F1 on average. Kang et al. (2023) compared MoCo v2, SwAV, Barlow Twins and DINO pre-trained on 19 million patches from 20,994 TCGA slides, and domain-aligned pre-training outperformed ImageNet pre-training in both linear and fine-tuning evaluation. Foundation models extend this further:

- CTransPath, a convolutional network combined with a multi-scale Swin Transformer (Wang et al., 2022);
- Phikon, an iBOT ViT-Base trained on more than 40 million images from 16 cancer types (Filiot et al., 2023);
- UNI, trained on more than 100 million images from over 100,000 whole-slide images (WSIs) covering 20 tissue types (Chen et al., 2024);

- CONCH, trained on more than 1.17 million image–caption pairs (Lu et al., 2024);
- Virchow, a 632-million-parameter ViT trained with DINOv2 on about 1.5 million WSIs (Vorontsov et al., 2024);
- Prov-GigaPath, trained on 1.3 billion tiles from 171,189 slides from a network of 28 cancer centers (Xu et al., 2024).

Scale does not by itself remove center effects. Kömen et al. (2024) found hospital signatures in foundation-model embeddings that dominated feature-space distances and were not removed by stain normalization. All ten public foundation models evaluated by de Jong et al. (2025) encoded the medical center strongly. Only one had a robustness index above one, meaning that biological features outweighed center features, and then only slightly. The PathoROB benchmark found robustness deficits in all 20 models it evaluated (Kömen et al., 2026). More robust models, vision–language alignment and post-hoc robustification reduced, but did not eliminate, the risk of the resulting downstream errors. Our analysis is complementary. We measure how the downstream performance of frozen and fine-tuned pathology encoders (Kang et al., 2023) varies across held-out domains, under the same protocol and diagnostics as every other arm.

2.5. Test-time adaptation

AdaBN adapts a trained network without labels by replacing its batch-normalization (BN) statistics with target-domain statistics, and it adds no parameters (Li et al., 2018). Schneider et al. (2020) showed that re-estimating BN statistics on corrupted images improved ImageNet-C robustness for 25 architectures, and argued that adapted results should be reported in OOD evaluations. Prediction-time BN improved accuracy and calibration under covariate shift, but its results were mixed when combined with pre-training and weaker under more natural shifts (Nado et al., 2020). TENT (Wang et al., 2021) minimizes prediction entropy online, updating normalization statistics and channel-wise affine parameters on each batch. Because these methods estimate statistics from test batches, which depend on batch composition, our protocol shuffles the test stream with a fixed seed before any adaptation.

2.6. Variance, rankings and statistical reporting

Data sampling, parameter initialization and hyperparameter choice each have a marked effect on benchmark results (Bouthillier et al., 2021), and test-set scores alone do not settle which model is best (Dodge et al., 2019). In biomedical image analysis challenges, an algorithm’s rank is generally not robust to the test data, the ranking scheme or the annotators (Maier-Hein et al., 2018). challengeR provides bootstrap analyses of ranking stability (Wiesenfarth et al., 2021). Varoquaux (2018) showed that cross-validation error bars are large (about $\pm 10\%$ with 100 samples) and that the standard error across folds strongly underestimates them. In two Kaggle medical-imaging challenges with test sets below 1,000 cases, the difference between the public and private leaderboards was larger than the gap between the winner and the top-10% entry (Varoquaux and Cheplygina, 2022). Any variance estimator computed only from the results of a cross-validation experiment is biased, and ignoring training-set variability can make an algorithm look significantly better than it is (Nadeau and Bengio, 2003). Cross-validation estimates the average error of models fit to other training sets from the same population (proved for least-squares linear models), and the correlation between folds makes the usual intervals too narrow (Bates et al., 2024). Leave-one-domain-out designs are an extreme case. There are few folds ($K = 3$ to 7 here), the folds share training domains, and the held-out domain is both a nuisance factor and the quantity of interest. We therefore report between-domain and between-run variance components with intervals and run a fold-dependence sensitivity analysis on paired contrasts. We also give bootstrap intervals for method ranks (over seeds and over domains) and for Kendall’s τ between cohort rankings.

3. Evaluation protocol

A cohort has K domains, and fold t holds out domain t for testing. Every arm (a training method or a test-time procedure) is run in every fold with several seeds. The cohorts, models and arms are described in Section 4.

3.1. Leave-one-domain-out design

Leave-one-domain-out evaluation is standard in domain generalization (Gulrajani and Lopez-Paz, 2021), and Camelyon17-WILDS adds a held-out validation hospital (Koh et al., 2021). We follow both. On Camelyon17, the canine cohort and MIDOG++, fold t also holds out domain $(t - 1) \bmod K$ as an out-of-distribution (OOD) validation

domain and trains on the remaining $K - 2$ domains. MIDOG 2021 has three labeled scanners, so each fold trains on two and has no OOD validation domain. Every domain is the test domain of exactly one fold, so the evaluation is *domain-complete*: no single test domain is privileged. In every training domain, an in-distribution (ID) validation split is held out that is disjoint from training in its groups (patients, slides or images; Section 4.1). The study adds four elements to this design.

Seed replication Every arm is run in every fold with five seeds at the compact tier and three at the standard tier (Section 4.2). The seed fixes the initialization of randomly initialized layers, the batch order, the augmentation draws and the order of the test stream for test-time adaptation. All arms use the same seed values.

Fixed step budget Every fine-tuned arm is trained for 1,200 steps with batch size 128 and a cosine learning-rate schedule. It is evaluated at ten checkpoints, at steps $\text{round}(kn/10)$ for $k = 1, \dots, 10$ and a budget of n steps. The only exception is the compute-matched control (Section 4.4), which needs a larger budget by design. With a fixed budget, schedule length and early stopping cannot confound method comparisons.

All model-selection rules At each checkpoint we evaluate the ID validation, OOD validation and test splits. Four rules select a checkpoint by the maximum area under the ROC curve (AUROC) on their selection split, with ties going to the earliest checkpoint:

1. *training-domain (ID) validation*, the primary rule (DomainBed’s default; Gulrajani and Lopez-Paz, 2021), available in every cohort;
2. *leave-one-domain-out validation* on the OOD validation domain (Camelyon17, canine, MIDOG++);
3. *last*, the final checkpoint;
4. *oracle*, the best test AUROC, reported only as an upper bound.

All four rules use the same ten checkpoints of the same runs, so any difference between them comes from selection alone.

Per-fold hyperparameter selection Hyperparameters are tuned with seed 0 on the grids of Table 2. Every fold keeps the configuration with the highest ID-validation AUROC at the ID-selected checkpoint of its own tuning run. Diverged runs are excluded unless every configuration diverged, and test data are never used. We do not average ID-validation AUROC over folds, because doing so leaks test-domain labels. The test domain of fold t is a training domain of the other folds, except for fold $t + 1$ when a validation domain exists. The ID-validation splits of those folds therefore contain labeled patches of domain t . In every cohort, some of these patches are pixel-identical to rows of fold t ’s test set. Averaging would thus let fold t ’s test labels influence the hyperparameters reported for fold t . At the standard tier, the base learning rate is selected first, and the method grids are then run at each fold’s selected rate. The Barlow Twins fine-tunes have their own rate grid (Section 4.3).

3.2. Reported quantities

Let y_{ts} be the test AUROC (the primary metric) of one arm in fold t with seed $s = 1, \dots, n_t$ under a given selection rule, and $\bar{y}_t = n_t^{-1} \sum_s y_{ts}$. For every arm, cohort, tier and rule we report

$$\text{worst domain: } \min_t \bar{y}_t, \quad (1)$$

$$\text{across-domain SD: } s_{\bar{y}} = \left[\frac{1}{K-1} \sum_t (\bar{y}_t - \hat{\mu})^2 \right]^{1/2}, \quad (2)$$

$$\text{mean: } \hat{\mu} = \frac{1}{K} \sum_t \bar{y}_t, \quad \hat{\mu} \pm t_{0.975, K-1} s_{\bar{y}} / \sqrt{K}. \quad (3)$$

With balanced seeds, the interval in (3) is exact under model (5), since $\text{Var}(\bar{y}_t) = \sigma_D^2 + \sigma_R^2/n$ for every domain. Percentile bootstrap intervals (Efron and Tibshirani, 1993) are supplementary only, because the domain bootstrap under-covers for $K \leq 7$. Calibration is reported as the expected calibration error (Pakdaman Naeini et al., 2015; Guo et al., 2017), with 15 equal-width bins B_b on the confidence of the predicted class (decision threshold 0.5):

$$\text{ECE} = \sum_{b=1}^{15} \frac{|B_b|}{N_{\text{test}}} \left| \text{acc}(B_b) - \text{conf}(B_b) \right|, \quad (4)$$

summarized by its mean and maximum over domains.

3.3. Variance decomposition

For one arm on one cohort we fit the one-way random-effects model

$$y_{ts} = \mu + b_t + e_{ts}, \quad b_t \sim \mathcal{N}(0, \sigma_D^2), \quad e_{ts} \sim \mathcal{N}(0, \sigma_R^2), \quad s = 1, \dots, n_t, \quad (5)$$

with all terms independent. Seeds are nested within domains: seed s in folds t and t' denotes different runs on different training sets, so no crossed seed effect is fitted. σ_D is the between-domain and σ_R the between-run SD. With $N = \sum_t n_t$, between- and within-domain mean squares $\text{MSB} = \sum_t n_t (\bar{y}_t - \bar{y})^2 / (K - 1)$ and $\text{MSW} = \sum_{t,s} (y_{ts} - \bar{y}_t)^2 / (N - K)$, and $n_0 = (N - \sum_t n_t^2 / N) / (K - 1)$ ($n_0 = n$ for balanced data), the ANOVA estimators (Searle et al., 1992) are

$$\hat{\sigma}_R^2 = \text{MSW}, \quad \hat{\sigma}_D^2 = \max\{(\text{MSB} - \text{MSW}) / n_0, 0\}, \quad \widehat{\text{ICC}} = \frac{\text{MSB} - \text{MSW}}{\text{MSB} + (n_0 - 1) \text{MSW}}, \quad (6)$$

where $\text{ICC} = \sigma_D^2 / (\sigma_D^2 + \sigma_R^2)$ is ICC(1) (Shrout and Fleiss, 1979; McGraw and Wong, 1996), reported truncated at 0. Restricted maximum likelihood (REML) estimates (Patterson and Thompson, 1971) are obtained by profiling out σ_R^2 and minimizing over $\gamma = \sigma_D^2 / \sigma_R^2 \geq 0$, with the boundary $\gamma = 0$ checked explicitly. They coincide with ANOVA for balanced data with a positive estimate.

Intervals and boundary estimates We report the exact F -based 95% interval for the ICC, the exact χ^2 interval for σ_R and the modified large-sample interval for σ_D (Searle et al., 1992; McGraw and Wong, 1996; Burdick and Graybill, 1992). With $K \leq 7$, $\hat{\sigma}_D$ is often at or near zero, so we also report a Bayesian grid posterior under weakly informative half-normal and half-Cauchy priors (scale 0.1 AUROC) (Gelman, 2006), with one-sided intervals $[0, q_{0.95}]$ at the boundary. The formulas are in Supplementary Section S11.

The ICC as a diagnostic The ICC is the share of an arm's single-run variance that is due to which domain is held out. Near one, the choice of domains dominates. Near zero, run-to-run noise is as large as the differences between domains, and a single-seed result says little about a domain. The ICC is conditional on arm, cohort, metric, selection rule and the domains at hand, and with $K \leq 7$ its intervals are wide. It tells us where the variance of a reported number comes from. It is not a quality standard, and we set no threshold for it. As an external reference point we apply the same estimators to DomainBed's published per-environment results (Gulrajani and Lopez-Paz, 2021). The mean $\pm$ standard error over three trials is converted to an SD with the convention of DomainBed's released code (Supplementary Section S10).

3.4. Paired contrasts and fold dependence

Each arm is compared with ERM of the same tier and cohort through $d_t = \bar{y}_t^a - \bar{y}_t^{\text{ERM}}$. We report $\bar{d}$, the interval $\bar{d} \pm t_{0.975, K-1} s_d / \sqrt{K}$, the two-sided p -value and the number of improved domains. The p -values are Holm-adjusted over all arms compared with ERM within a cohort, tier and selection rule (Holm, 1979).

The folds of a leave-one-domain-out design share training domains, so the d_t are not independent (Nadeau and Bengio, 2003; Bates et al., 2024). Let T_i be the set of training domains of fold i and $O_{ij} = |T_i \cap T_j| / \sqrt{|T_i| |T_j|}$ the overlap of folds i and j . As a sensitivity analysis we assume a working correlation $\text{corr}(d_i, d_j) = R_{ij} = \rho O_{ij}$ for $i \neq j$ ($R_{ii} = 1$). Under this assumption $\text{Var}(\bar{d}) = \sigma^2 \mathbf{1}^\top \mathbf{R} \mathbf{1} / K^2$, and the sample variance is biased downward, $\mathbb{E}[s_d^2] = \sigma^2(1 - \bar{r})$, where $\bar{r}$ is the mean off-diagonal element of $\mathbf{R}$. The corrected standard error and the effective number of folds are then

$$\widehat{\text{SE}}_\rho(\bar{d}) = \left[\frac{s_d^2}{1 - \bar{r}} \frac{\mathbf{1}^\top \mathbf{R} \mathbf{1}}{K^2} \right]^{1/2}, \quad K_{\text{eff}} = \frac{K^2}{\sum_{i,j} R_{ij}}, \quad (7)$$

with t_{K-1} quantiles, in the spirit of the corrected resampled t statistic of Nadeau and Bengio (2003). Because ρ cannot be estimated from K folds, we report $\rho \in \{0, 0.25, 0.5, 1\}$. For our designs, $K_{\text{eff}} = 3.33, 2.50$ and 1.67 at $\rho = 0.25, 0.5$ and 1 for Camelyon17 and the canine cohort ($K = 5$); $3.50, 2.33$ and 1.40 for MIDOG++ ($K = 7$); and $2.40, 2.00$ and 1.50 for MIDOG 2021 ($K = 3$).

3.5. Power

Power for a paired comparison depends on the variance of the paired difference, not on either arm's between-domain SD. Writing $y_{ts}^a = \mu_a + b_t + (ab)_t + e_{ts}^a$ and $y_{ts}^{\text{ERM}} = \mu_{\text{ERM}} + b_t + e_{ts}^{\text{ERM}}$, the domain effect b_t cancels, and with s seeds

per arm

$$\sigma_{\Delta}^2(s) = \text{Var}(d_t) = \sigma_{DA}^2 + \sigma_{R,\Delta}^2/s, \quad \sigma_{R,\Delta}^2 = \text{Var}(e_{ts}^a - e_{ts}^{\text{ERM}}), \quad (8)$$

where σ_{DA}^2 is the domain-by-arm interaction variance. Because all arms share seed values, their run noise may be correlated, so $\sigma_{R,\Delta}^2$ is estimated as the pooled within-domain variance of the per-seed differences $y_{ts}^a - y_{ts}^{\text{ERM}}$, which is unbiased either way, and $\hat{\sigma}_{DA}^2 = \max\{s_d^2 - \hat{\sigma}_{R,\Delta}^2 h, 0\}$ with $h = K^{-1} \sum_t n_t^{-1}$. If seed sets are not aligned, we use $\hat{\sigma}_{R,\Delta}^2 = \text{MSW}_a + \text{MSW}_{\text{ERM}}$ and $\hat{\sigma}_{DA}^2 = \max\{s_d^2 - \text{MSW}_a h_a - \text{MSW}_{\text{ERM}} h_{\text{ERM}}, 0\}$. The power of the two-sided paired t test over K domains to detect a mean difference δ is

$$1 - \beta(K) = \Pr(|T_{K-1}(\lambda)| > t_{1-\alpha/2, K-1}), \quad \lambda = \frac{\delta \sqrt{K_{\text{eff}}}}{\sigma_{\Delta}(s)}, \quad (9)$$

where $T_{K-1}(\lambda)$ is noncentral t with $K - 1$ degrees of freedom and $K_{\text{eff}} = K$ for independent folds. We report the smallest K that reaches 80% power at $\alpha = 0.05$, for $\delta \in \{0.02, 0.05, 0.10, 0.20\}$ AUROC and $s \in \{1, 3, 5, 10\}$. For $\rho > 0$, σ_{Δ} is first divided by $\sqrt{1 - \bar{r}_{\text{obs}}}$ (the bias correction of (7) for the observed design). The primary *ratio* mode then holds the observed inflation K/K_{eff} fixed as K grows. The *design* mode uses K_{eff} of a hypothetical K -domain design, which levels off near $1/\rho$ because training sets overlap more as K grows, so some targets are unreachable.

3.6. Rank stability and the task-by-shift decomposition

Within a cohort, arms are ranked by worst-domain (1) and by mean AUROC, with 95% percentile rank intervals and the probability of ranking first from 5,000 bootstrap replicates. Two resampling schemes are used: seeds within each (arm, domain) cell, and domains with one shared draw for all arms. Agreement between two cohorts' rankings is Kendall's τ_b over their common arms (Kendall, 1938), with a 95% interval from 2,000 seed-bootstrap replicates and a p -value against independent orderings. Of the six cohort pairs, two share the task, two the shift type and two neither. We report the mean worst-domain τ_b per class and bootstrap intervals of the differences, which are descriptive.

Arm-by-cohort interaction For the A arms that are run in all four cohorts, let $G_{cta} = \bar{y}_{ct}^a - \bar{y}_{ct}^{\text{ERM}}$ be the gain of arm a over ERM in domain t of cohort c . Each (cohort, domain) row is centered across arms, $E_{cta} = G_{cta} - A^{-1} \sum_{a'} G_{cta'}$, removing the domain's average gain. Whether the centered profiles differ between cohorts is tested with domains as replicates, using a studentized Welch–James statistic and a wild bootstrap that multiplies whole residual rows by Rademacher signs (10,000 replicates). This keeps each cohort's variance and the correlation between arms.

Task, shift and task $\times$ shift Task and shift type are constant within a cohort, so they can only be assessed with the cohort as the unit. Let $\mathbf{m}_c \in \mathbb{R}^A$ be the mean of the centered rows of cohort c , with equal weight for every cohort, and let $s_c^f = \pm 1$ code cohort c for factor $f \in \{\text{task}, \text{shift}, \text{task} \times \text{shift}\}$, with $s_c^{\text{task} \times \text{shift}} = s_c^{\text{task}} s_c^{\text{shift}}$. Because the four cohorts form a 2×2 design with one cohort per cell, the between-cohort interaction sum of squares splits exactly into three orthogonal contrasts:

$$\sum_c \|\mathbf{m}_c - \bar{\mathbf{m}}\|^2 = \sum_f \text{SS}_f, \quad \text{SS}_f = \|\mathbf{L}_f\|^2, \quad \mathbf{L}_f = \frac{1}{2} \sum_c s_c^f \mathbf{m}_c. \quad (10)$$

We report each SS_f and its fraction of the total, and the rank of the task and shift splits among the three possible two-versus-two splits of four cohorts. A cohort-level permutation p -value equals rank/3 and cannot be below $1/3$. We also give an approximate test $F_f = \text{SS}_f / \text{SS}_{\text{task} \times \text{shift}}$ on $(A - 1, A - 1)$ degrees of freedom for $f \in \{\text{task}, \text{shift}\}$. It treats the cohort-level deviations as independent and spherical across arms. Its error term has only $A - 1$ degrees of freedom and absorbs any genuine task-by-shift pattern. With four cohorts, this decomposition describes the interaction. It cannot show that rankings are organized by task or by shift type, and we do not claim that they are.

3.7. A site acceptance test for new sites

The diagnostics above describe a method across sites. A practitioner who is about to use one trained model at a new hospital or scanner needs a decision for that site: is the model's AUROC there at least a target T ? We propose an empirical-Bayes site acceptance test that combines the development evidence with a small labeled sample from the new site.

Development prior Let $z_{sr} = \text{logit}(y_{sr})$ be the test AUROC of run r of the method at development site s on the logit scale, for K' held-out development sites with n runs each. Under model (5) on this scale, with a flat prior on μ and independent half-Cauchy priors on σ_D and σ_R , the AUROC of one run at a new site has, on the logit scale, the posterior predictive distribution

$$\theta_{\text{new}} \mid \sigma_D, \sigma_R, \mathbf{z} \sim \mathcal{N}\left(\bar{z}, \sigma_D^2 + \sigma_R^2 + \frac{\sigma_D^2 + \sigma_R^2/n}{K'}\right), \quad (11)$$

integrated over the grid posterior of (σ_D, σ_R) , where $\bar{z}$ is the grand mean. The result is a scale mixture of normal distributions centered at $\bar{z}$. Its spread grows with the between-site SD and shrinks only slowly with K' , so few or heterogeneous development sites give a vague prior. The half-Cauchy scale is 0.5 on the logit scale (about 0.06 AUROC at an AUROC of 0.85).

Local evidence At the new site, m patches are labeled. Their AUROC a_m (with a continuity correction that keeps it inside $(0, 1)$) gives $\hat{z}_m = \text{logit}(a_m)$ with variance $v_m = \text{Var}(a_m)/\{a_m(1 - a_m)\}^2$, where $\text{Var}(a_m)$ is the Hanley–McNeil variance for the observed numbers of positives and negatives. Each mixture component is updated conjugately with $\hat{z}_m \sim \mathcal{N}(\theta, v_m)$, and the component weights are updated by their marginal likelihoods.

Decision The site is accepted if $P(\theta \geq \text{logit } T \mid \mathbf{z}, \hat{z}_m) \geq 1 - \alpha$, with $\alpha = 0.05$. The posterior also gives a point estimate and a credible interval for the site's AUROC. Before any labels are collected, the number of labels needed is planned by pre-posterior simulation: sites are drawn from the development prior (11), local scores are simulated, and m is the smallest sample size for which sites that exceed the target by a chosen margin (0.02) are accepted with the chosen probability (0.8). The simulation uses equal-variance binormal scores at a prevalence of 0.5.

Validation design Every held-out site of every cohort plays the new site in turn, for every method and seed (the six extra frozen backbones of the representation ladder excepted). Its prior is built only from the other folds' results of the same method at their own held-out sites, so the new site's test labels are never used. Samples that contain only one class are discarded (at most 3 of 200). Labeled samples of size $m \in \{25, 50, 100, 200, 400\}$ are drawn at random, 200 times, from the run's saved test predictions, and the truth is the run's AUROC on the full test split. We compare the test with two baselines: local labels alone (accept if the one-sided 95% Wald bound of $\hat{z}_m$ exceeds $\text{logit } T$) and the development prior alone ($m = 0$). We report the rate of accepting a site whose true AUROC is below T (false acceptance) and the rate of accepting a site at or above T (power), for $T \in \{0.80, 0.85, 0.90\}$.

4. Materials and experimental setup

4.1. Cohorts

The four primary cohorts form a 2×2 design that crosses the task (tumor versus non-tumor tissue, or mitotic figure versus non-mitotic look-alike) with the shift type. Under an *institutional* shift, domains differ in center, staining, scanner and patients together. Under an *acquisition-only* shift, the domains are scanners within one laboratory (Fig. 1, Table 1). All splits are row-shuffled with a fixed seed. Within a fold, the labeled splits are pairwise disjoint in their groups (patient, slide or image), and the unlabeled pool shares no group with the validation or test splits. A verification script recomputes these conditions from the stored group identifiers. It also checks that every labeled split contains both classes, that ID validation covers every training domain and that no patch is duplicated within or across splits. Every fold passes.

Cohort construction Camelyon17-WILDS provides 96×96 px lymph-node patches from five hospitals (Koh et al., 2021; Bandi et al., 2019); ID validation holds out whole patients of every training hospital. In the canine cohort, each of 44 slides was digitized with five scanners (Wilm et al., 2023). The slides are split into five groups, so that a test domain is a new scanner and a set of slides that is absent from training under every scanner, and magnification is normalized to the CS2 scanner. MIDOG 2021 (Aubreville et al., 2023a) has 150 annotated breast cancer images from one laboratory, 50 per scanner. The scanners' resolutions differ by about 11%, so the primary cache normalizes every image to 0.25 $\mu\text{m}/\text{px}$; the un-normalized cache is a compact-tier sensitivity analysis. MIDOG++ (Aubreville et al., 2023b) provides 503 images of seven tumor types from two species, and its 150 breast cancer images are the MIDOG 2021 images. Mitosis patches are centered on annotated mitotic figures and non-mitotic look-alikes.

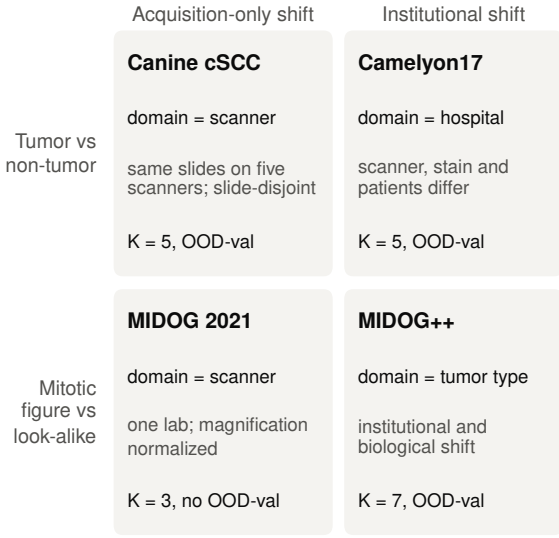
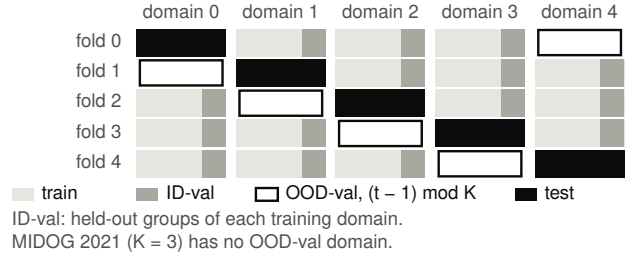
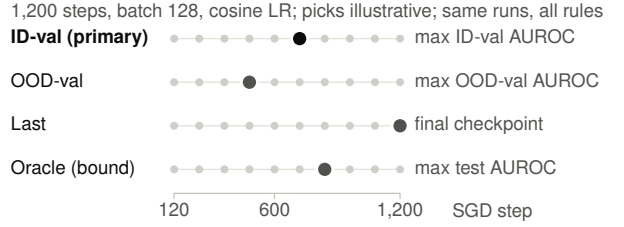
a Task × shift-type design (one cohort per cell)

b Leave-one-domain-out folds (K = 5 shown)

c Ten checkpoints per run, four selection rules


Figure 1: Study design. (a) The four cohorts cross the task with the type of shift, one cohort per cell. (b) Leave-one-domain-out folds for $K = 5$: fold t tests domain t , validates out of distribution on domain $(t - 1) \bmod K$ and holds out ID-validation groups of every training domain. (c) Every run is checkpointed ten times and evaluated under four selection rules; the positions of the selected checkpoints are illustrative.

Table 1

Cohorts and per-fold split sizes (number of patches; a range is the minimum–maximum over folds). Counts are identical at 64 and 96 px. ID-val: group-disjoint validation split of the training domains; OOD-val: held-out validation domain $(t - 1) \bmod K$. The un-normalized MIDOG 2021 variant, used as a compact-tier sensitivity analysis, has the same split sizes as MIDOG 2021.

	Camelyon17-WILDS	Canine cSCC	MIDOG 2021	MIDOG++
Task (pos. vs neg.)	tumor vs normal (lymph node)	SCC tumor vs non-tumor tissue	mitotic figure vs look-alike	mitotic figure vs look-alike
Domain	hospital	scanner (same 44 slides)	scanner (one lab)	tumor type
K	5	5	3	7
Shift type	institutional	acquisition-only	acquisition-only	institutional and biological
Disjoint group	patient	slide	image	image
OOD-val domain	yes	yes	none	yes
Scale normalization	none (native patch)	per scanner, to CS2	0.25 $\mu\text{m}/\text{px}$	0.25 $\mu\text{m}/\text{px}$
Train	30,000	22,872–26,596	2,160–2,544	11,854–17,940
ID-val	12,000	3,895–4,780	597–716	2,555–3,891
OOD-val	12,000	2,197–3,745	–	1,806–8,216
Test	25,000	2,197–3,745	1,175–1,648	1,806–8,216
Unlabeled pool	40,000	40,000	25,200	40,000
Positive rate, train	0.50	0.52–0.55	0.35–0.39	0.42–0.49
Positive rate, test	0.50	0.51–0.59	0.35–0.43	0.27–0.63

The unlabeled pools used for self-supervision come only from the training domains and exclude all validation and test groups, so the self-supervised arms remain domain generalization methods. Sampling, scale normalization, the MIDOG++ domains and the handling of objects near image borders are detailed in Supplementary Section S12.

4.2. Model tiers

Compact tier (C) The compact model is a four-stage convolutional network on 64 px input with 583,394 parameters. Stages 1–3 each apply two 3×3 convolution–batch normalization–ReLU blocks followed by 2×2 max pooling, with 32,

64 and 128 channels. Stage 4 is one such block with 256 channels, followed by global average pooling and a linear layer with two outputs. It is trained from random initialization, and inputs are normalized with ImageNet channel statistics.

Standard tier (S) ResNet-50 (He et al., 2016) with ImageNet initialization (Deng et al., 2009) is fine-tuned end to end at the native 96 px. Two pathology foundation models of Kang et al. (2023) are also used: a Barlow Twins ResNet-50, fine-tuned at 96 px, and a DINO ViT-S/16 (Dosovitskiy et al., 2021), fine-tuned on inputs upsampled bilinearly to 224 px. Their inputs are normalized with the channel statistics released with the weights. Frozen linear probes on ImageNet ResNet-50, Barlow Twins ResNet-50 and DINO ViT-S/16 follow one rule: the input is upsampled to 224 px, and features are extracted in fp32 and standardized with the training-split mean and SD. A linear head is trained with SGD (learning rate 0.05, momentum 0.9, weight decay 10^{-4} , 1,200 steps of 128) and selected like every other arm. A supplementary representation ladder applies the same probe to nine frozen backbones. Kang et al. (2023) pre-trained on 19 million patches from 20,994 TCGA whole-slide images, and the released checkpoints are these TCGA-only models. None of our cohorts comes from TCGA. Camelyon17 comes from Dutch hospitals, the canine slides from the veterinary CATCH collection (Wilm et al., 2023), and MIDOG++ from veterinary laboratories and UMC Utrecht. The pre-training images therefore do not overlap with ours, although some organs (breast, skin) occur in both.

4.3. Optimization and checkpointing

Training uses the fixed budget of Section 3.1: batches of 128 drawn without replacement and reshuffled every epoch, and a cosine schedule. The compact tier uses SGD (momentum 0.9, weight decay 10^{-4}) with a fixed, untuned learning rate of 0.02. The standard-tier ResNet-50 arms use SGD with the same momentum and weight decay. Their learning rate is selected per fold from $\{10^{-3}, 3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}, 10^{-1}\}$ on ImageNet ERM and shared by every ImageNet ResNet-50 arm of the fold. The grid was extended upward twice because the selected rate sat on its upper edge. The Barlow Twins fine-tunes have their own per-fold rate, selected on Barlow Twins ERM from $\{3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}\}$. Inheriting the ImageNet rate made their training fail on some folds (Supplementary Section S8). The ViT uses AdamW (weight decay 0.05, 5% linear warm-up) with a learning rate selected per fold from $\{10^{-5}, 3 \times 10^{-5}, 10^{-4}\}$. Every arm except the frozen probes receives a random dihedral augmentation (flips and rotations by multiples of 90° , drawn per batch). The standard tier trains in bfloat16 autograd and the compact tier in fp32. Evaluation is always in fp32, with float64 probabilities. TF32 is enabled, and kernels are not forced to be deterministic. At each checkpoint the record stores eight metrics per split (AUROC, average precision, accuracy, balanced accuracy, F1, ECE, Brier score, negative log-likelihood) and the ID-validation AUROC of each training domain. Predictions at the selected checkpoints and the ID-selected weights are also saved. A run whose loss or predictions become non-finite is stopped, marked as diverged and kept in the analysis, with non-finite probabilities scored as 0.5. Six of the 3,400 final runs diverged, all IRM: two at the compact tier and four at the standard tier. Camelyon17 models are also scored on the PatchCamelyon test split (Veeling et al., 2018), center-cropped to 64 px at the compact tier. This score is never used for selection.

4.4. Arms

Table 2 lists every arm with its implementation and grid. Arms that combine two components inherit the tuned value of their parent: SimCLR+SAM takes ρ from SAM, and the foundation-model H&E-jitter arms take s from standard-tier H&E jitter.

Controls for self-supervision The augmentation-only control applies SimCLR’s view augmentation to labeled batches without the contrastive loss, which separates the effect of the augmentations from that of pre-training. The compute-matched control gives ERM as many images through the encoder as SimCLR receives in pre-training and fine-tuning together (Table 2).

Stain normalization As in common practice, Macenko’s stain matrix (Macenko et al., 2009) is estimated once per slide imaged on one scanner, from the pooled tissue pixels of up to 256 of its tiles, with concentrations clamped at zero, and is mapped to a reference fitted with the same estimator on the training slides. No labels are used, but the method is transductive: each test slide is normalized from its own tiles. The per-tile variant estimates the stain matrix on every tile separately. Implementation details are in Supplementary Section S12, and the effect of the granularity in Supplementary Section S9. Our H&E jitter perturbs RGB color; it is not an augmentation in HED space.

Table 2

Arms, implementations and hyperparameter grids. Grids are searched per fold with seed 0 and selected by ID-validation AUROC; values outside a grid are fixed. Tier: C = compact CNN (64 px), S = standard (96 px). All fine-tuned arms also receive the shared dihedral augmentation. [†]Transductive: uses the unlabeled images of the test domain.

Arm	Implementation	Grid / fixed values	Tier
<i>Baseline and controls</i>			
ERM	Cross-entropy.	C: lr 0.02; S: lr $\in \{10^{-3}, 3 \times 10^{-3}, \dots, 10^{-1}\}$	C, S
Augmentation-only	SimCLR's view augmentation (random resized crop, flips, rotations, color jitter, grayscale) applied to supervised batches; no contrastive loss.	crop scale [0.4, 1]; jitter 0.4 with prob. 0.8; hue offset 0.08; grayscale prob. 0.2	C
Compute-matched ERM	ERM for 1,200 + $\lceil I_{SSL}/128 \rceil$ steps; I_{SSL} = images passed through the encoder during SimCLR pre-training (two views per image).	4,944 steps ($ U = 40,000$); 3,552 steps (MIDOG 2021, $ U = 25,200$)	C
<i>Optimization and self-supervision</i>			
SAM (Foret et al., 2021)	SGD base optimizer; batch-norm statistics updated on the unperturbed pass only.	$\rho \in \{0.02, 0.05, 0.1\}$	C, S
SimCLR (Chen et al., 2020)	NT-Xent pre-training on the fold's unlabeled pool (C: from scratch; S: continued from ImageNet), 2-layer projection head, then ERM fine-tuning.	$\tau = 0.5$, batch 256, 6 epochs, lr 0.05 (C) / 0.01 (S)	C, S
SimCLR+SAM	SimCLR initialization, SAM fine-tuning.	ρ from SAM	C
<i>Domain-aware training (uses training-domain labels)</i>			
GroupDRO (Sagawa et al., 2020)	Weights over training domains, $q_k \leftarrow q_k e^{\eta L_k}$, loss $\sum_k q_k L_k$; domains absent from a batch are not updated.	$\eta \in \{10^{-3}, 10^{-2}, 10^{-1}\}$	C, S
IRMv1 (Arjovsky et al., 2019)	Split-batch penalty per domain, averaged over domains; weight 1 for the first 25% of steps, then λ , with optimizer reset.	$\lambda \in \{1, 10, 100\}$	C, S
DeepCORAL (Sun and Saenko, 2016)	Squared differences of feature means and covariances, averaged over pairs of training domains.	$\lambda \in \{0.1, 1, 10\}$	C, S
MixStyle (Zhou et al., 2021)	Random mixing of instance feature statistics after stages 1 and 2 (C) or ResNet layers 1 and 2 (S); identity at evaluation.	$p \in \{0.5, 1\} \times \alpha \in \{0.1, 0.3\}$	C, S
Fish (Shi et al., 2022)	Inner copy takes one SGD step per training domain in the batch (random order, persistent inner optimizer); $\theta \leftarrow \theta + \epsilon(\tilde{\theta} - \theta)$ for parameters, batch-norm buffers copied.	$\epsilon \in \{0.01, 0.05, 0.1, 0.5\}$	C, S
LISA (Yao et al., 2022)	Per batch, intra-label mixup across domains (prob. p_{sel}) or intra-domain mixup across labels; $\lambda \sim \text{Beta}(\alpha, \alpha)$, soft labels.	$\alpha \in \{0.5, 2\} \times p_{sel} \in \{0.5, 1\}$	C, S
<i>Stain normalization and color augmentation</i>			
Macenko [†] (Macenko et al., 2009)	One stain estimate per slide and scanner, mapped to a reference fitted with the same estimator on the training slides; every split.	$I_0 = 240$, OD threshold 0.15, angle percentile 1; ≤ 256 tiles per slide	C, S
H&E jitter	Per-image brightness, contrast and saturation factors $\sim U[1-s, 1+s]$ and per-channel offsets $\sim U[-s/5, s/5]$.	$s \in \{0.2, 0.4\}$	C, S
TIA-style [†]	Per-slide Macenko normalization followed by H&E jitter.	$s = 0.3$	C, S
Per-tile Macenko	One stain estimate per tile (Supplementary Section S9).	as Macenko	C, S
<i>Test-time adaptation (on the ID-selected checkpoint)</i>			
BN-adapt [†] (Schneider et al., 2020)	Batch-norm statistics re-estimated (cumulative average) on the shuffled test stream in batches of 128, then prediction.	on ERM, SimCLR, H&E jitter (C); ERM (S)	C, S
TENT [†] (Wang et al., 2021)	Online entropy minimization of batch-norm affine parameters (Adam, lr 10^{-3} , one step per batch of 128, shuffled stream); a batch is predicted before its update.	on ERM	C
<i>Pathology foundation models (Kang et al., 2023)</i>			
Barlow Twins	Fine-tuned at 96 px.	lr $\in \{3 \times 10^{-4}, \dots, 3 \times 10^{-2}\}$ on BT ERM; s from H&E jitter	S
RN50: ERM, +H&E jitter			
DINO ViT-S/16: ERM, +H&E jitter	Fine-tuned at 224 px, AdamW.	lr $\in \{10^{-5}, 3 \times 10^{-5}, 10^{-4}\}$; s from H&E jitter	S
Frozen probes	Linear head on frozen ImageNet RN50, Barlow Twins RN50 or DINO ViT-S/16 features at 224 px.	lr 0.05	S

Test-time adaptation BN-adapt and TENT process the test split in a seed-specific random order. Each batch of 128 therefore contains both classes in roughly the test domain's proportion. A label-sorted stream would give single-class batches and an uninformative adapted AUROC.

Transductive arms BN-adapt, TENT, Macenko and TIA-style use the unlabeled images of the test domain at prediction time; the other arms see one test image at a time. The four transductive arms are marked in every table and figure and are not directly comparable with the inductive arms.

Dropped arm The rotation-invariance arm of the previous version was dropped. In this pipeline it reduced to the dihedral augmentation that every arm receives, so it was identical to ERM.

4.5. Seeds, compute and availability

Final runs use seeds 0–4 at the compact tier and 0–2 at the standard tier. Tuning uses seed 0 and is stored separately from the final runs. The planned design has the following runs:

- tuning: 520 at the compact tier (26 configurations, 20 folds) plus 78 for the un-normalized MIDOG 2021 cohort; at the standard tier, 160 learning-rate runs (100 ResNet-50, 60 ViT), 100 Barlow Twins learning-rate runs and 520 method-grid runs;
- final: 1,600 at the compact tier (16 arms, 5 seeds, 20 folds), 240 for the un-normalized MIDOG 2021 sensitivity analysis, and 1,200 at the standard tier (20 arms, 3 seeds, 20 folds);
- representation ladder: 360 runs.

Test-time adaptation is evaluated within the run of the base arm. All 4,778 runs (1,378 tuning and 3,400 final, the ladder included) completed. They ran on NVIDIA H100 NVL GPUs and used about 34 GPU-hours, not counting development and diagnostic runs. Each record contains the full hyperparameters, a SHA-256 hash of the training code and a fingerprint of the fold cache. A record whose fingerprint or hyperparameters do not match the job is recomputed. Code, fold-construction scripts, all run records and the scripts that regenerate every table and figure are available at <https://github.com/InvenTide-AI/histopath-shift> [TBD: Zenodo DOI of the archived release].

5. Results

No model is trained on labeled data from its test domain. Apart from the test-time adaptation arms, which use unlabeled test images, every result below is the performance of a model applied unchanged to an unseen site. Unless stated otherwise, results use the primary model-selection rule (training-domain validation) and test AUROC. We lead with the standard tier, whose models are closer to current practice, and give the compact tier alongside. Arms marked [†] use unlabeled test-domain images and are not like-for-like competitors of the other arms (Section 4.4).

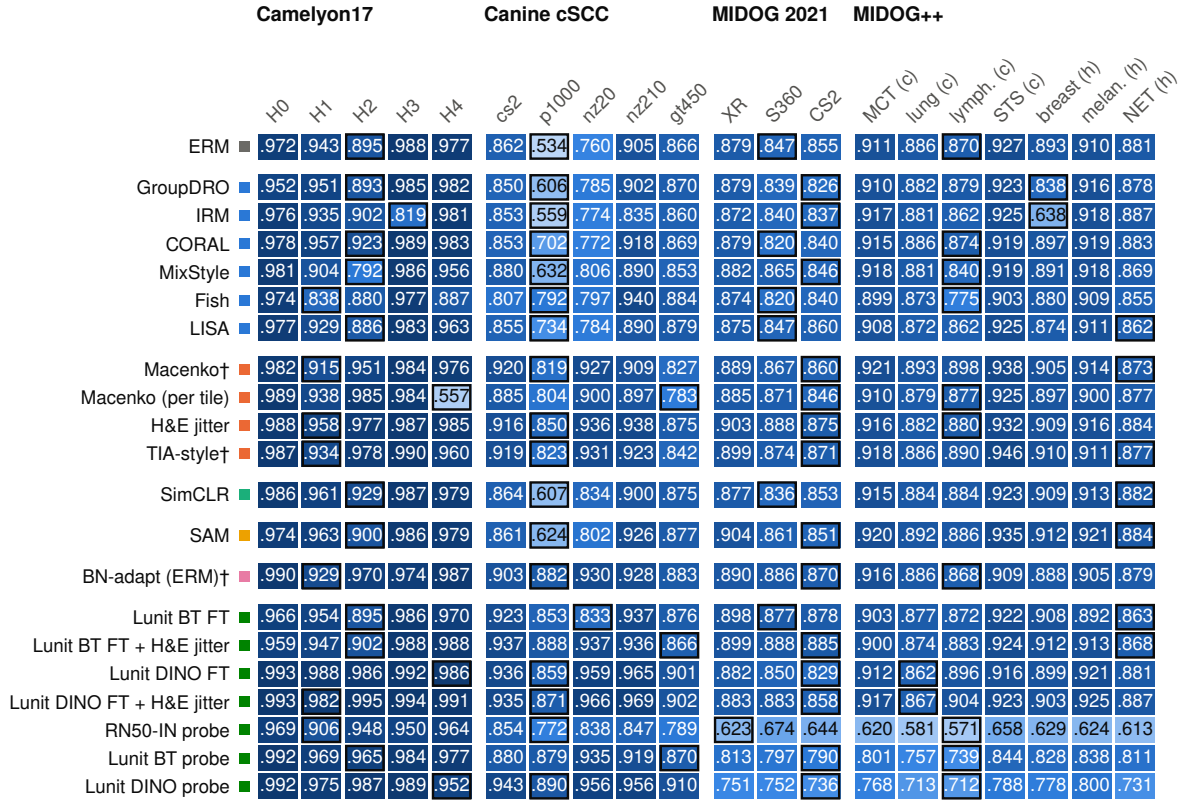
5.1. Performance at every held-out domain

Figure 2 shows the seed-mean test AUROC of every standard-tier arm at every held-out domain, and Table 3 summarizes it by the worst domain and the mean. Averages hide large differences between domains. Standard-tier ERM has a mean AUROC of 0.785 on the canine cohort but scores 0.534 on the 3DHistech P1000 scanner, which is close to chance. That scanner is the worst domain for 17 of the 21 arms. On Camelyon17, ERM averages 0.955 and drops to 0.895 at center 2. The two mitosis cohorts are flatter: ERM ranges from 0.847 to 0.879 on MIDOG 2021 and from 0.870 to 0.927 on MIDOG++. The compact tier shows the same pattern at a lower level (Supplementary Table S4). There, ERM's worst domains score 0.698 (Camelyon17), 0.551 (canine), 0.765 (MIDOG 2021) and 0.766 (MIDOG++).

The worst domain is not the same for every arm. On Camelyon17, center 2 is the worst domain for 10 of 21 standard-tier arms, and the others fail at centers 1, 3 or 4. A study that reports a single held-out hospital therefore measures one point of a distribution whose shape depends on the method.

5.2. Where the variance of a held-out score comes from

Figure 3 and Table 4 report the variance decomposition of Section 3.3. For most arms, most of the variance of a single run is due to which domain is held out. The median ICC over arms lies between 0.72 and 0.95 in every cohort



Outlined: worst domain of the arm (rows with every domain present).

■ Reference ■ Self-supervision ■ Test-time adaptation
 ■ DG objectives ■ Sharpness ■ Pretrained representations
 ■ Stain & color

0.5 0.6 0.7 0.8 0.9 1.0
test AUROC (seed mean, ID-val selection)

† Transductive: uses the unlabeled images of the test domain at prediction time.

Figure 2: Test AUROC of every standard-tier arm at every held-out domain (mean over three seeds; ID-validation selection). Outlined cells mark each arm's worst domain. Domains: Camelyon17 hospitals H0–H4; canine scanners; MIDOG 2021 scanners; MIDOG++ tumor types ((c) canine, (h) human). †Transductive: uses the unlabeled images of the test domain at prediction time. The compact tier is in Supplementary Fig. S2.

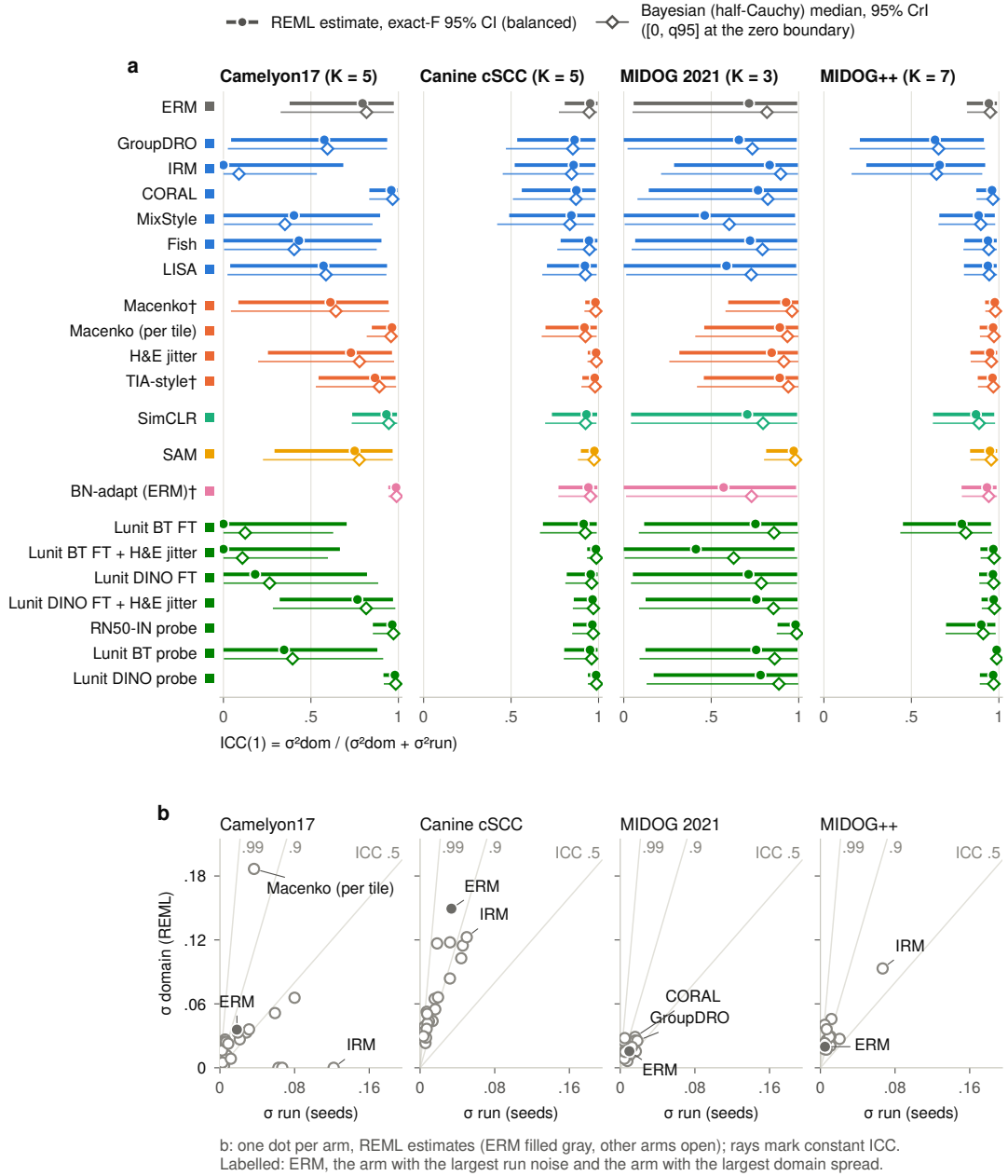
and at both tiers. It is highest on the canine cohort and MIDOG++ at the standard tier (0.95 in both), where 20 of 21 and 18 of 21 arms have an ICC interval whose lower limit exceeds 0.5. For comparison, the same estimators give a median ICC of 0.98 on DomainBed's published results outside TerraIncognita (Supplementary Section S10).

Run-to-run noise is nevertheless far from negligible, and it is largest at the hard domains. At the compact tier, the five ERM seeds at Camelyon17 center 2 span 0.599 to 0.929 AUROC, and at center 4, ERM's worst domain, they span 0.592 to 0.797 (Supplementary Fig. S6). At the standard tier, the three ERM seeds on the canine P1000 scanner span 0.461 to 0.598. For some arms the spread at the worst domain is larger still: the seeds of standard-tier IRM at Camelyon17 center 3 span 0.50 to 0.98, and those of compact-tier per-tile Macenko at center 4 span 0.41 to 0.74.

The intervals are wide. On MIDOG 2021, with $K = 3$, only 3 of 21 standard-tier arms have an ICC lower limit above 0.5, and for five arms the lower limit is zero. Estimates at the zero boundary occur for three standard-tier arms on Camelyon17, and the Bayesian posterior gives them one-sided intervals (Fig. 3a). The ICC is therefore best read as a coarse description of each cohort, not as a property of a method.

5.3. Can the differences between methods be detected?

Table 4 counts the paired comparisons with ERM. At the standard tier, no arm is significantly better than ERM after Holm adjustment in any cohort. The only Holm-significant differences are three frozen linear probes that are worse



† Transductive: uses the unlabeled images of the test domain at prediction time.

Figure 3: Variance decomposition at the standard tier. (a) ICC(1), the share of single-run variance due to the held-out domain, with the REML estimate and exact- F 95% interval and the Bayesian (half-Cauchy) posterior median and 95% credible interval; one-sided $[0, q_{0.95}]$ at the zero boundary. (b) Between-domain SD $\hat{\sigma}_D$ against between-run SD $\hat{\sigma}_R$ per arm; rays mark constant ICC. † Transductive. The compact tier is in Supplementary Fig. S3.

than ERM on MIDOG++. At the compact tier, two arms improve on ERM after adjustment, both on MIDOG++: compute-matched ERM (+0.060) and per-slide Macenko† (+0.041). The augmentation-only control, LISA and IRM are significantly worse there.

Lack of significance does not mean that the differences are small. On the canine cohort, the Lunit DINO frozen probe improves mean AUROC over ERM by 0.145 and is better at all five scanners, yet its unadjusted p is 0.071. With

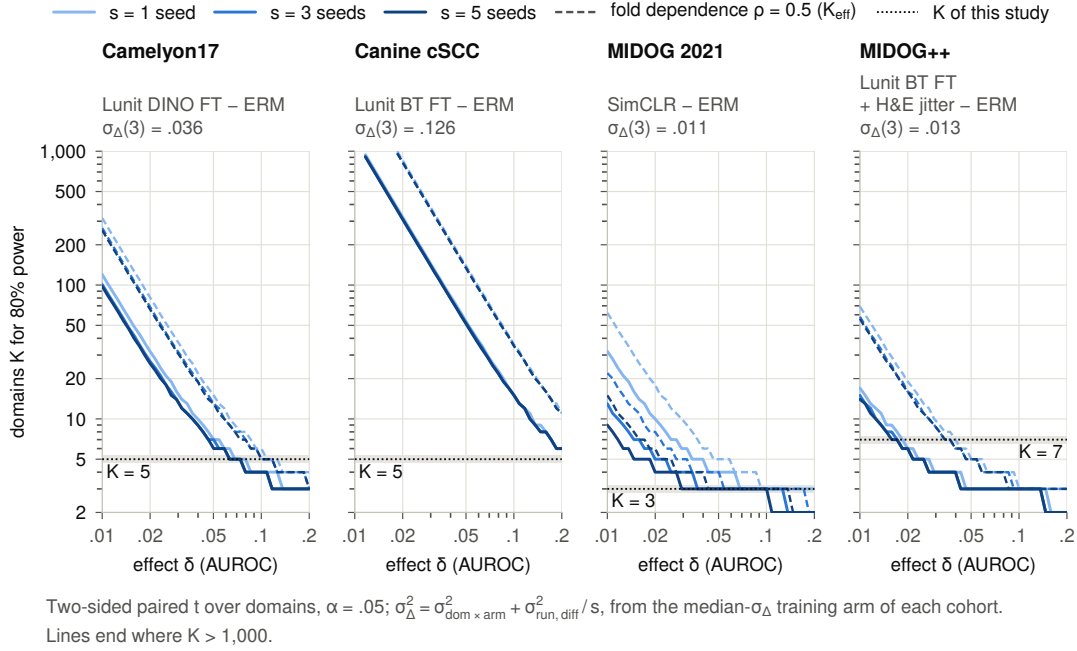


Figure 4: Number of held-out domains K needed for 80% power to detect a mean AUROC gain δ over ERM (two-sided paired t test, $\alpha = 0.05$), at the standard tier, for the training arm with the median σ_{Δ} in each cohort. Solid: independent folds; dashed: fold dependence $\rho = 0.5$; dotted: K of this study. Colors: seeds per arm. The compact tier is in Supplementary Fig. S4.

five domains, even an exact sign-flip test cannot go below 0.0625 when every domain improves. On the tumor cohorts the gains are large but vary strongly from domain to domain. On the mitosis cohorts the reverse holds: the differences are measured precisely but are small. The largest standard-tier gain on MIDOG++ is +0.010 (SAM, better at all seven domains, Holm $p = 0.053$).

The power analysis puts numbers on this (Fig. 4, Table 4). The SD of the per-domain difference from ERM, $\hat{\sigma}_{\Delta}$, has a median of 0.012–0.028 on the mitosis cohorts but 0.036–0.126 on the tumor cohorts. For a median arm to detect a mean gain of 0.05 AUROC with 80% power, the standard tier would need 7 Camelyon17 hospitals and 53 canine scanners. The compact tier would need 30 and 36. If the folds are correlated through their shared training domains ($\rho = 0.5$), these numbers roughly double or triple. On the mitosis cohorts, 3–5 domains suffice for a gain of that size, but the observed gains there are much smaller. Adding seeds helps little once σ_{Δ} is dominated by the domain-by-arm interaction (Fig. 4). Domains, not seeds, are the scarce resource.

5.4. Rankings within a cohort, and the effect of model scale

Figure 5a gives each arm’s worst-domain rank with bootstrap intervals over seeds and over domains. Resampling domains usually widens the intervals more than resampling seeds. The median width of the domain-bootstrap interval is 3 to 7 ranks out of 21 at the standard tier, against 2 to 6 for the seed bootstrap. Only one standard-tier arm has the rank interval $[1, 1]$ under both bootstraps: the Lunit DINO probe on the canine cohort, which ranks first in 99% of domain resamples. The next most certain is fine-tuned Barlow Twins with H&E jitter on MIDOG 2021 (96%, interval $[1, 3]$). On MIDOG++, the top arm (SAM) ranks first in only 45% of domain resamples.

A few patterns hold across cohorts.

- *Pathology foundation models lead on the tumor cohorts.* The fine-tuned Lunit DINO ViT ranks first on Camelyon17 (worst domain 0.986), and its frozen probe ranks first on the canine cohort (0.890). On the mitosis cohorts, the frozen probes fall to the bottom: the DINO probe ranks 20th and 19th, and the ImageNet ResNet-50 probe last on both. Fine-tuning narrows the gap: fine-tuned Barlow Twins with H&E jitter ranks first on MIDOG 2021 (0.885), and the fine-tuned DINO ViT ranks 15th and 13th on the two mitosis cohorts.

- *H&E color jitter is consistently good.* It ranks 4th, 7th, 3rd and 3rd at the standard tier and 3rd, 8th, 5th and 8th at the compact tier.
- *The domain-generalization objectives rarely help.* Under ID-validation selection, none of the six (GroupDRO, IRM, CORAL, MixStyle, Fish and LISA) ranks in the top five on any cohort at either tier. The best of them is CORAL on MIDOG++ (6th at the standard tier). Other rules can lift a single objective: with oracle selection, CORAL ranks 4th on MIDOG++ at the standard tier and LISA 2nd on the canine cohort at the compact tier.
- *Transductive arms are strong.* BN-adapt on ERM ranks 2nd on the canine cohort, and at the compact tier BN-adapt on H&E jitter ranks among the top six everywhere. Both use unlabeled test-domain images.

Model scale changes the picture (Fig. 6). Moving from the compact CNN to ImageNet ResNet-50 raises ERM's worst-domain AUROC on Camelyon17 (0.698 to 0.895) and on the mitosis cohorts (0.765 to 0.847; 0.766 to 0.870), but not on the canine cohort (0.551 to 0.534). For the 14 arms run at both tiers, Kendall's τ_b between the two tiers' worst-domain rankings is 0.60 (Camelyon17), 0.63 (canine), 0.32 (MIDOG 2021) and 0.30 (MIDOG++). Conclusions drawn from the compact model therefore transfer only in part. Compute-matched ERM, the best compact arm on MIDOG++, was not run at the standard tier, so that result cannot be checked there.

Hyperparameters matter as much as methods. The Barlow Twins fine-tunes first inherited the learning rate tuned for ImageNet ResNet-50, and training failed on some folds. Tuning their own rate raised mean off-site AUROC on MIDOG++ from 0.715 to 0.891 (Supplementary Section S8.1). Stain normalization shows the same sensitivity to implementation. Estimating the Macenko stain matrix per tile instead of per slide drops the arm from rank 10 to last on Camelyon17 at the standard tier, because it fails at one center (0.557 against 0.976 for the per-slide estimate; Supplementary Section S9).

5.5. Do rankings follow the task or the type of shift?

The arm-by-cohort interaction is significant at both tiers (wild bootstrap, $p < 0.001$ at the standard tier and $p = 0.004$ at the compact tier): methods do not rank the same way on every cohort. Figure 7 shows how the rankings agree between pairs of cohorts.

At the standard tier, agreement follows the task. The two tumor cohorts agree with $\tau_b = 0.43$ (95% interval [0.28, 0.52]) and the two mitosis cohorts with $\tau_b = 0.42$ [0.30, 0.50]. The pairs that share only the shift type agree much less (0.15 and 0.14), and the pairs that share neither do not agree at all (0.03 and -0.06). The mean τ_b of same-task pairs exceeds that of same-shift pairs by 0.28 [0.19, 0.40]. In the 2×2 decomposition of Eq. (10), the task contrast accounts for 84% of the between-cohort sum of squares, the shift contrast for 10% and their interaction for 7%. The pattern is the same under the last-checkpoint (85% task) and oracle (90%) rules.

At the compact tier the picture is less clear. The task contrast accounts for 51% and the shift contrast for 32%, and same-task pairs agree only slightly more than same-shift pairs (difference 0.06 [-0.04, 0.14]). The canine and MIDOG 2021 cohorts, which share the shift type but not the task, agree with $\tau_b = 0.54$, almost as much as the two mitosis cohorts (0.59).

Two cautions apply. First, with four cohorts there are only three ways to split them into two pairs, and the task split ranks first among them at both tiers. A permutation test at the cohort level cannot go below $p = 1/3$. The decomposition describes the interaction; it cannot prove that rankings are organized by task. Second, the 150 breast-cancer images of MIDOG++ are the MIDOG 2021 images, so the two mitosis cohorts share one of MIDOG++'s seven domains. This overlap may raise their agreement. The tumor cohorts share no images, and their agreement at the standard tier is as high.

5.6. Sensitivity analyses

Model-selection rule Table 5 compares the worst-domain ranking under each rule with the ID-validation ranking. At the standard tier, the rules agree closely (τ_b 0.83–0.99), and the top arm is the same under every rule on Camelyon17 and MIDOG++. At the compact tier, agreement is weaker where run noise is large. On the canine cohort, τ_b is 0.38 for OOD-validation selection and 0.32 for the oracle, and on Camelyon17 it is 0.53–0.63. A compact-tier ranking thus depends on the selection rule as well as on the seeds (Supplementary Fig. S8).

Scale normalization of MIDOG 2021 Without normalizing the $\approx 11\%$ difference in scanner magnification, the compact-tier ranking of 20 arms changes little ($\tau_b = 0.78$ for the worst domain and 0.83 for the mean). No arm's worst-domain AUROC moves by more than 0.039 (Supplementary Fig. S12).

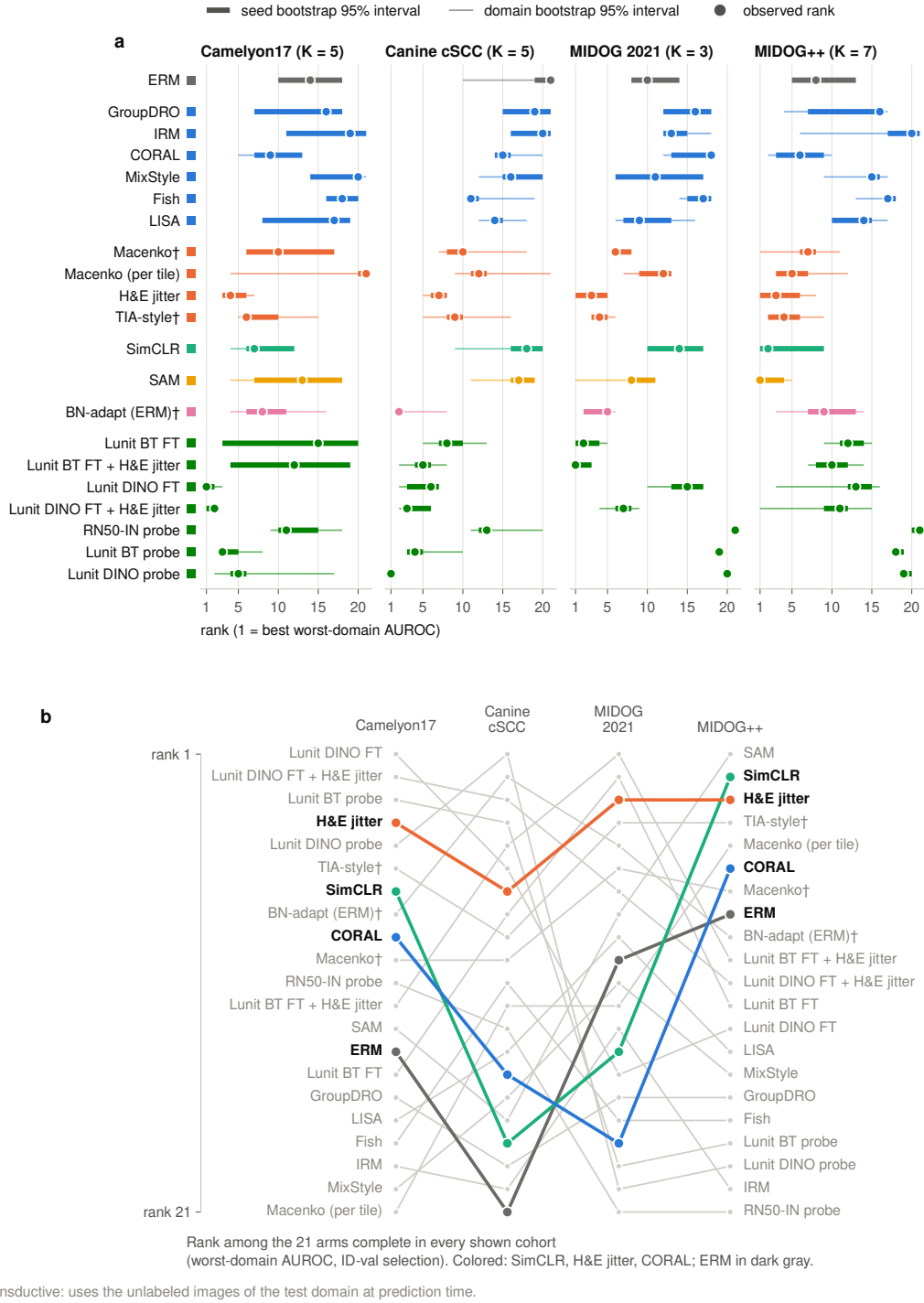


Figure 5: Rankings at the standard tier. (a) Rank by worst-domain AUROC (1 = best) with 95% intervals from bootstrapping seeds within domains (thick) and domains (thin). (b) Ranks of all 21 arms across the four cohorts; ERM, SimCLR, H&E jitter and CORAL are highlighted. † Transductive. The compact tier is in Supplementary Fig. S5.

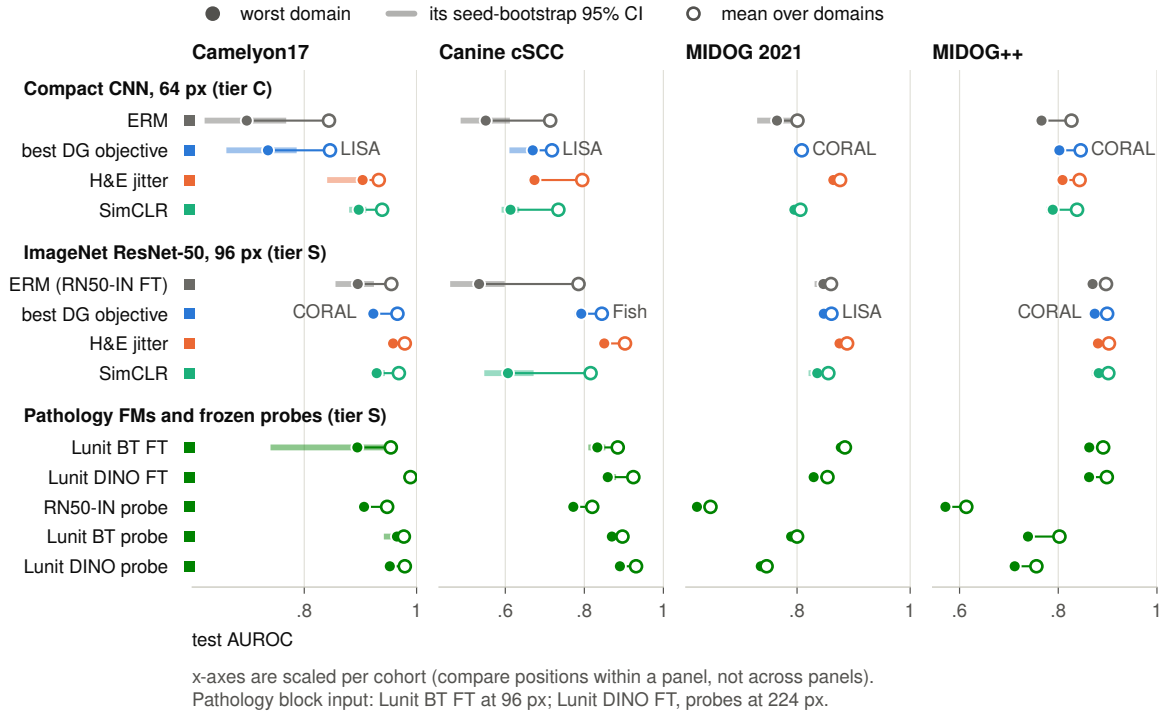


Figure 6: Effect of model scale and pre-training. Worst-domain AUROC (filled, with seed-bootstrap 95% interval) and mean AUROC (open) for ERM, the best domain-generalization objective, H&E jitter and SimCLR at the compact tier (compact CNN, 64 px) and the standard tier (ImageNet ResNet-50, 96 px), and for the pathology foundation models. The x-axes are scaled per cohort.

PatchCamelyon Camelyon17 models were also scored on the PatchCamelyon test split. PatchCamelyon comes from Camelyon16, whose two hospitals also contribute to Camelyon17 (Veeling et al., 2018; Litjens et al., 2018), so it is a secondary check, not an independent cohort. At the standard tier the Lunit models, fine-tuned or frozen, score highest, led by fine-tuned DINO with H&E jitter (mean over folds 0.908, against 0.772 for ERM). At the compact tier TIA-style (0.802) and the augmentation arms score highest, against 0.698 for ERM (Supplementary Table S5).

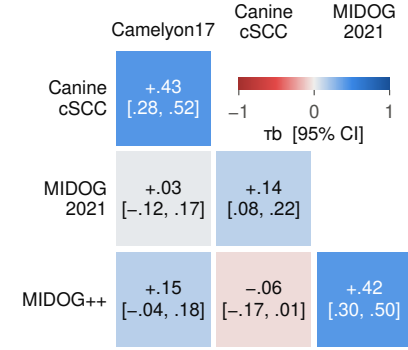
Calibration On every cohort at the standard tier, one of the two fine-tuned foundation models with H&E jitter has the lowest mean ECE (0.027–0.095), against 0.089–0.180 for ERM (Supplementary Fig. S9). Per-slide normalization with jitter (TIA-style) and jitter alone are close: on the canine cohort, their ECE is 0.075 and 0.080 at the standard tier and 0.047 and 0.065 at the compact tier.

Frozen representations Among nine frozen backbones under one linear-probe protocol (Supplementary Fig. S11), the Lunit Barlow Twins ResNet-50 has the best worst-domain AUROC averaged over cohorts (0.841), followed by the Lunit DINO ViT-S/16 (0.822). Pathology pre-training is not enough by itself, however. The Lunit SwAV (0.707) and MoCo v2 (0.734) ResNet-50s are below ImageNet ConvNeXt-T (0.780) and ViT-B/16 (0.774), and SwAV is below even the ImageNet ResNet-50 (0.718).

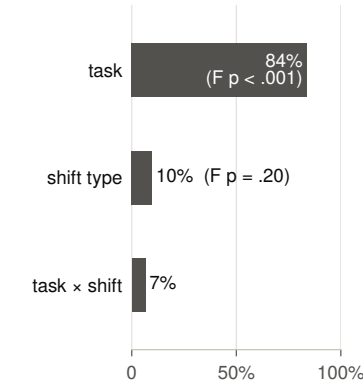
5.7. The site acceptance test

Figure 8 and Table 6 report the validation of the site acceptance test of Section 3.7 over the 1,200 standard-tier runs, with every held-out site playing the new site in turn. The development prior alone is not a safe basis for a decision. Without local labels it accepted 8.6% of the sites whose true AUROC was below 0.80, because a model that works at the development sites can still fail at a new one.

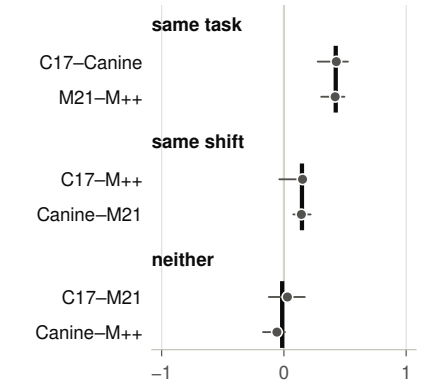
With local labels, the test kept false acceptance at or below 2.1% for every target and sample size, and at or below 1.3% for sites more than 0.02 AUROC below the target. Its 90% credible intervals covered the true site AUROC in

a Kendall τ_b between cohort rankings


Arms ranked by worst-domain AUROC;
21 common complete arms per pair.

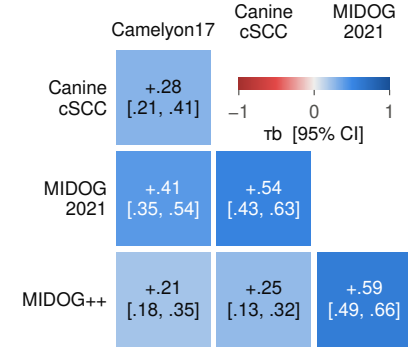
b Split of between-cohort SS


Per-domain Δ AUROC vs ERM of the
20 other arms common to all 4 cohorts.
Cohort-level split descriptive
(3 balanced 2-vs-2 splits).

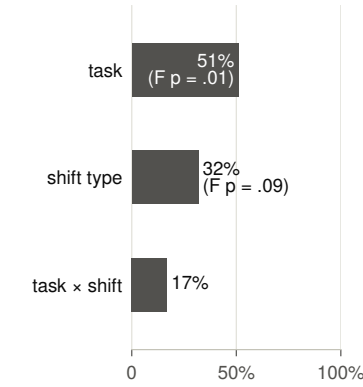
c τ_b by cohort-pair type


same task – same shift:
+.28 [+.19, +.40]

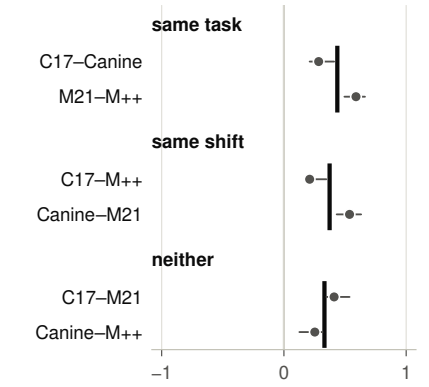
● pair τ_b , 95% CI — class mean

a Kendall τ_b between cohort rankings


Arms ranked by worst-domain AUROC;
20 common complete arms per pair.

b Split of between-cohort SS


Per-domain Δ AUROC vs ERM of the
19 other arms common to all 4 cohorts.
Cohort-level split descriptive
(3 balanced 2-vs-2 splits).

c τ_b by cohort-pair type


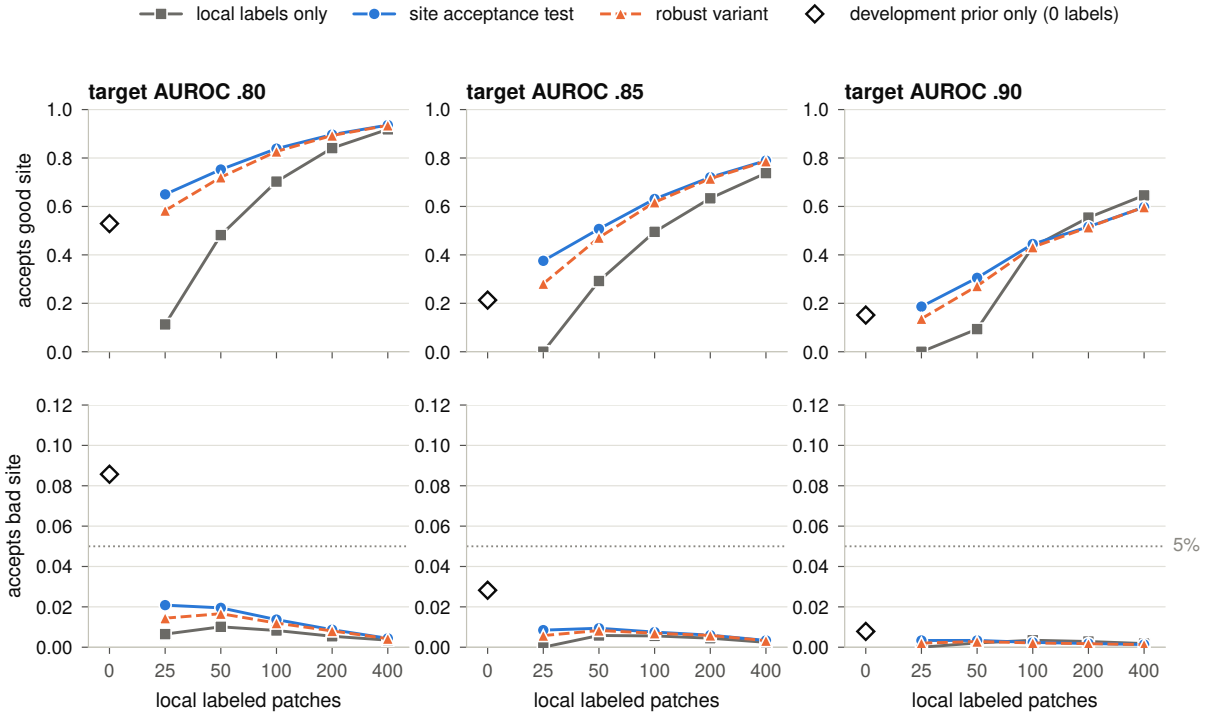
same task – same shift:
+.06 [-.04, +.14]

● pair τ_b , 95% CI — class mean

Figure 7: Agreement of rankings between cohorts at the standard tier (top) and the compact tier (bottom). (a) Kendall's τ_b of worst-domain rankings with 95% seed-bootstrap intervals. (b) Split of the between-cohort sum of squares of the per-domain gains over ERM into task, shift type and their interaction (Eq. 10); descriptive, since only three two-versus-two splits exist. (c) τ_b by type of cohort pair.

92–93% of cases. For targets of 0.80 and 0.85 it needed about half as many labels as local labels alone. At a target of 0.80, it accepted 75% of the good sites with 50 labels and 84% with 100, while local labels alone accepted 48% and 70% and needed 200 labels to reach 84%. At a target of 0.85, the test reached with 100 labels (63%) what local labels alone reached with 200. Its point estimates were also more accurate, with a mean absolute error of 0.024 AUROC at 50 labels against 0.036. The price is a slightly higher false-acceptance rate than local labels alone, by about one percentage point at the lower targets. The gain is largest where the development sites score well above the target, as on Camelyon17 and MIDOG++. Where they score near or below it, as on the canine cohort and at the compact tier, the prior is pessimistic and can make the test less powerful than local labels alone (Supplementary Table S3).

The advantage has limits. At the strict target of 0.90 with 200 or more labels, local labels alone were slightly more permissive (55% against 52% accepted at 200 labels). In the hardest subgroup, Camelyon17 at a target of 0.90, where most runs score above 0.95 and the 24 runs below the target mostly fall just short of it, false acceptance reached 8.7%



1200 runs of the standard tier; each held-out site plays the new site, with the prior built from the other folds only. Labeled samples drawn at random from the site's test patches (200 draws per run and size). Accept if $P(\text{AUROC} \geq \text{target}) \geq 0.95$ (local only: one-sided 95% bound $\geq \text{target}$).

Figure 8: Validation of the site acceptance test at the standard tier. Top: share of good sites (true AUROC $\geq$ target) that each rule accepts; bottom: share of bad sites (true AUROC $<$ target) that it accepts, with the 5% level dotted. Means over runs of the acceptance rate over 200 random labeled samples per run and sample size. The compact tier is in Supplementary Fig. S1.

with 25 labels and 5.7% with 50, and fell to 3.3% at 100 and 1.7% at 200. An ϵ -contaminated prior, which gives 10% of the prior mass to a vague component centered at an AUROC of 0.5, lowered these values to 5.4%, 4.5% and 2.8%, at a cost in power at small samples (for example from 38% to 28% at 25 labels and a target of 0.85; Supplementary Section S1). We added this variant after seeing the subgroup result and report it as a sensitivity analysis. At the compact tier, whose sites score lower and differ more, the test was as safe (false acceptance at most 0.4%), but it gained over local labels alone only at small samples and lost some power at larger ones. The decision threshold $1 - \alpha$ is a posterior probability, so these false-acceptance rates are empirical averages over the mix of bad sites in the study, not a guaranteed error rate; for runs just below the target (within 0.02) false acceptance was higher, up to 5.8% (target 0.80, 50 labels). The label planner was close to the observed acceptance at the compact tier and slightly optimistic at the standard tier, where it predicted 75% acceptance of good sites at 100 labels against 68% observed.

Table 3

Standard tier: worst-domain AUROC (rank among the 21 arms) and mean AUROC over held-out domains, ID-validation selection, three seeds. Bold: best worst-domain AUROC in the cohort. [†]Transductive. FT: fine-tuned; BT: Barlow Twins; IN: ImageNet. The compact tier is in Supplementary Table S4.

Arm	Camelyon17 ($K = 5$)		Canine cSCC ($K = 5$)		MIDOG 2021 ($K = 3$)		MIDOG++ ($K = 7$)	
	worst (rank)	mean	worst (rank)	mean	worst (rank)	mean	worst (rank)	mean
<i>Reference and controls</i>								
ERM	0.895 (14)	0.955	0.534 (21)	0.785	0.847 (10)	0.860	0.870 (8)	0.897
<i>Domain-aware training</i>								
GroupDRO	0.893 (16)	0.953	0.606 (19)	0.803	0.826 (16)	0.848	0.838 (16)	0.890
IRM	0.819 (19)	0.923	0.559 (20)	0.776	0.837 (13)	0.850	0.638 (20)	0.861
CORAL	0.923 (9)	0.966	0.702 (15)	0.823	0.820 (18)	0.846	0.874 (6)	0.899
MixStyle	0.792 (20)	0.924	0.632 (16)	0.812	0.846 (11)	0.864	0.840 (15)	0.891
Fish	0.838 (18)	0.911	0.792 (11)	0.844	0.820 (17)	0.845	0.775 (17)	0.871
LISA	0.886 (17)	0.948	0.734 (14)	0.828	0.847 (9)	0.861	0.862 (14)	0.888
<i>Stain normalization and color</i>								
Macenko [†]	0.915 (10)	0.962	0.819 (10)	0.881	0.860 (6)	0.872	0.873 (7)	0.906
Macenko (per tile)	0.557 (21)	0.891	0.783 (12)	0.854	0.846 (12)	0.867	0.877 (5)	0.895
H&E jitter	0.958 (4)	0.979	0.850 (7)	0.903	0.875 (3)	0.888	0.880 (3)	0.903
TIA-style [†]	0.934 (6)	0.970	0.823 (9)	0.887	0.871 (4)	0.881	0.877 (4)	0.905
<i>Self-supervision</i>								
SimCLR	0.929 (7)	0.968	0.607 (18)	0.816	0.836 (14)	0.855	0.882 (2)	0.901
<i>Sharpness-aware</i>								
SAM	0.900 (13)	0.960	0.624 (17)	0.818	0.851 (8)	0.872	0.884 (1)	0.907
<i>Test-time adaptation</i>								
BN-adapt (ERM) [†]	0.929 (8)	0.970	0.882 (2)	0.905	0.870 (5)	0.882	0.868 (9)	0.893
<i>Pretrained representations</i>								
Lunit BT FT	0.895 (15)	0.954	0.833 (8)	0.885	0.877 (2)	0.885	0.863 (12)	0.891
Lunit BT FT + H&E jitter	0.902 (12)	0.957	0.866 (5)	0.913	0.885 (1)	0.891	0.868 (10)	0.896
Lunit DINO FT	0.986 (1)	0.989	0.859 (6)	0.924	0.829 (15)	0.854	0.862 (13)	0.898
Lunit DINO FT + H&E jitter	0.982 (2)	0.991	0.871 (3)	0.929	0.858 (7)	0.875	0.867 (11)	0.904
RN50-IN probe	0.906 (11)	0.948	0.772 (13)	0.820	0.623 (21)	0.647	0.571 (21)	0.614
Lunit BT probe	0.965 (3)	0.977	0.870 (4)	0.896	0.790 (19)	0.800	0.739 (18)	0.802
Lunit DINO probe	0.952 (5)	0.979	0.890 (1)	0.931	0.736 (20)	0.746	0.712 (19)	0.756

Table 4

Summary of the evidence per tier and cohort (ID-validation selection). ICC: median and interquartile range over arms, and the number of arms whose exact 95% ICC interval has a lower limit above 0.5. ERM $\hat{\sigma}_D/\hat{\sigma}_R$: between-domain and between-run SD of ERM. $p < 0.05$ and Holm: number of arms whose paired mean difference from ERM is significant before and after Holm adjustment, split into better (+) and worse (−). Largest mean gain: the arm with the largest mean AUROC difference from ERM. $\hat{\sigma}_\Delta$: median over arms of the SD of the per-domain difference from ERM with the tier's seed count (Eq. 8). $K_{0.05}$: median over arms of the number of domains needed for 80% power to detect a mean gain of 0.05 AUROC, for independent folds and, in brackets, under fold dependence $\rho = 0.5$ (Eq. 9). Seeds: 3 (S), 5 (C). [†]Transductive.

Tier	Cohort	K	ICC median [IQR]	$CI_{lo} > 0.5$	ERM $\hat{\sigma}_D / \hat{\sigma}_R$	$p < 0.05$ (+/−)	Holm (+/−)	Largest gain over ERM	$\hat{\sigma}_\Delta$	$K_{0.05}$
S	C17	5	0.73 [0.40, 0.93]	7/21	0.036 / 0.018	0 / 0	0 / 0	Lunit DINO FT + H&E jitter (+0.036)	0.036	7 (14)
	Canine	5	0.95 [0.92, 0.98]	20/21	0.149 / 0.034	0 / 0	0 / 0	Lunit DINO probe (+0.145)	0.126	53 (137)
	MIDOG21	3	0.76 [0.71, 0.85]	3/21	0.016 / 0.010	4 / 3	0 / 0	Lunit BT FT + H&E jitter (+0.031)	0.013	3 (4)
	MIDOG++	7	0.95 [0.90, 0.97]	18/21	0.020 / 0.005	2 / 4	0 / 3	SAM (+0.010)	0.012	3 (5)
C	C17	5	0.81 [0.67, 0.89]	11/20	0.101 / 0.080	0 / 1	0 / 0	TIA-style [†] (+0.107)	0.094	30 (76)
	Canine	5	0.80 [0.68, 0.88]	10/20	0.103 / 0.055	0 / 0	0 / 0	BN-adapt (ERM) [†] (+0.123)	0.103	36 (92)
	MIDOG21	3	0.72 [0.38, 0.83]	3/20	0.037 / 0.025	7 / 0	0 / 0	BN-adapt (H&E jitter) [†] (+0.091)	0.028	5 (8)
	MIDOG++	7	0.85 [0.79, 0.92]	16/20	0.033 / 0.014	5 / 7	2 / 3	Compute-matched ERM (+0.060)	0.023	4 (10)

Table 5

Agreement between model-selection rules: Kendall's τ_b between the worst-domain ranking under each rule and the ranking under ID-validation selection, over the arms analyzed under every rule (TTA pseudo-arms are evaluated only under ID-validation selection). MIDOG 2021 has no OOD-validation domain. Same top arm: whether every rule puts the same arm first.

Tier	Cohort	Arms	τ_b OOD-val	τ_b last	τ_b oracle	Same top arm
S	Camelyon17	20	0.84	0.83	0.85	yes
	Canine cSCC	20	0.89	0.97	0.87	no
	MIDOG 2021	20	—	0.85	0.85	no
	MIDOG++	20	0.97	0.99	0.96	yes
C	Camelyon17	16	0.58	0.63	0.53	no
	Canine cSCC	16	0.38	0.90	0.32	yes
	MIDOG 2021	16	—	0.92	0.85	yes
	MIDOG++	16	1.00	1.00	0.98	yes

Table 6

Site acceptance test at the standard tier (1,200 runs). FA: false acceptance, the percentage of runs with true site AUROC below the target T that are accepted; power: the percentage of runs at or above T that are accepted; m : local labeled patches; MAE: mean absolute error of the point estimate of the site AUROC; cover.: coverage of the 90% interval. Rule: accept if $P(\text{AUROC} \geq T) \geq 0.95$ (local labels only: one-sided 95% Wald bound $\geq T$). The compact tier and the robust variant are in Supplementary Section S1.

Rule	m	$T = 0.80$		$T = 0.85$		$T = 0.90$		MAE	cover. (%)
		FA (%)	power (%)	FA (%)	power (%)	FA (%)	power (%)		
Development prior only	0	8.6	53	2.8	21	0.8	15	–	–
Site acceptance test	25	2.1	65	0.8	38	0.3	19	0.029	92
	50	2.0	75	0.9	51	0.3	31	0.024	92
	100	1.4	84	0.8	63	0.2	44	0.019	92
	200	0.9	90	0.6	72	0.2	52	0.015	92
	400	0.4	94	0.3	79	0.1	60	0.011	93
Local labels only	25	0.7	11	0.0	0	0.0	0	0.051	94
	50	1.0	48	0.6	29	0.2	9	0.036	94
	100	0.8	70	0.6	50	0.3	43	0.025	93
	200	0.5	84	0.4	63	0.3	55	0.017	93
	400	0.4	92	0.2	74	0.2	65	0.012	94

6. Discussion

6.1. What a domain-complete, seed-replicated evaluation shows

Four findings stand out.

First, a held-out score is mostly a statement about the held-out domain. For most arms, the domain explains most of the variance of a single run, and the same method can score near chance at one scanner and above 0.9 at another (Section 5.1). The worst domain also differs between methods. A result on one official test split therefore describes one domain, and a different split could reverse a comparison.

Second, with three to seven domains, most differences between methods cannot be resolved statistically. No standard-tier arm beats ERM after correcting for multiple comparisons, even when its mean gain is 0.145 AUROC and it improves every domain. On the tumor cohorts, gains are large but vary strongly between domains, so between 7 and 53 domains would be needed to confirm a gain of 0.05. On the mitosis cohorts, differences are measured precisely but are small. Adding seeds does not fix this, because the domain-by-method interaction dominates. This agrees with DomainBed, where no algorithm beat ERM by a margin the authors found significant (Gulrajani and Lopez-Paz, 2021), and with the large ERM seed variance reported for Camelyon17 in WILDS (Koh et al., 2021).

Third, rankings are uncertain within a cohort and change between cohorts. At the standard tier they agree more between cohorts that share a task than between cohorts that share a shift type. This reverses the thesis of our own earlier version, which held that the type of shift decides which method works best. That thesis rested on two cohorts that differed in task as well as in shift type, and the factorial design was built to separate the two. With four cohorts, one per cell, the design can describe the pattern but cannot prove it, and the compact tier does not clearly show it. We read the result as a warning, not a law. A method's rank on one task says little about its rank on another, even when the shift looks similar.

Fourth, implementation and tuning move results as much as the choice of method. Estimating the stain matrix per tile instead of per slide turned the fourth-best compact-tier arm on Camelyon17 into the worst. Inheriting a learning rate tuned for another initialization made Barlow Twins fine-tuning fail on MIDOG++. The defects that we found in our own earlier pipeline (Section 6.4) are of the same kind. A comparison of methods is only as good as the weakest implementation in it.

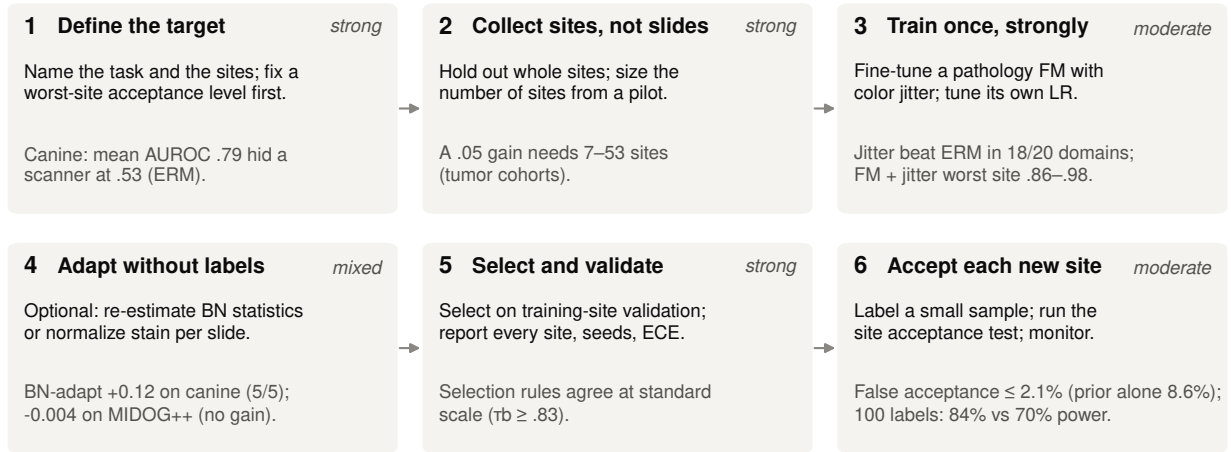
The method-level results are consistent with earlier work. Under the primary selection rule, none of the six domain-generalization objectives ranked in the top five on any cohort. This is consistent with DomainBed and WILDS, where such objectives did not reliably beat ERM (Gulrajani and Lopez-Paz, 2021; Koh et al., 2021). Color augmentation was consistently good, as Tellez et al. (2019) found for HSV and HED augmentation. Pathology foundation models were strong on the tumor cohorts, but their frozen features were weak for mitotic figures, and two of the four Lunit backbones were weaker frozen encoders than ImageNet ConvNeXt-T. Domain-aligned pre-training beats ImageNet pre-training on average (Kang et al., 2023), but which method helps varies across datasets and shifts (Wiles et al., 2022), and in our data it also varied with the task.

6.2. A roadmap for models that work across sites without retraining

Every number in this study answers a practical question: how well does a model trained once on some sites work at a site it has never seen? No model was trained on labeled data from its test site, and apart from the test-time adaptation arms, no model was changed at all. The study therefore doubles as a test of the “train once, deploy at many sites” strategy that practitioners want. At the standard scale, the best method of each cohort reached a worst-site AUROC of 0.884 to 0.986 without any retraining, against 0.534 to 0.895 for ERM. No single method was best everywhere, and no improvement over ERM could be confirmed statistically with the sites available. A practitioner can therefore improve the odds that a model transfers, and can find out whether it does, but cannot assume it.

Figure 9 and Table 7 turn the results into six steps. Each step carries an evidence grade. *Strong* means that the finding holds in all four cohorts at the standard tier. *Moderate* means a consistent direction that is not significant after correction, or a result that holds with exceptions. *Mixed* means large gains in some cohorts and none in others. Step 6 uses the site acceptance test proposed in Section 3.7.

Step 1: define the target before training Decide which task, which sites and which scanners the model is meant for, and fix the minimum performance that every site must reach. Judge models by their worst site. Averages hid a near-chance scanner on the canine cohort (mean AUROC 0.785, worst scanner 0.534 for ERM), and the hardest site differed between methods.



Evidence grades: strong = holds in all four cohorts at the standard tier; moderate = consistent direction but not significant after correction, or holding with exceptions; mixed = large gains in some cohorts, none in others. Numbers: standard tier, ID-val selection.

Figure 9: A six-step roadmap for pathology models that should work at new sites without retraining, with the evidence for each step from this study (standard tier, ID-validation selection) and its evidence grade. Details are in Table 7.

Step 2: collect sites, not only slides Gather development data from as many sites and scanners as possible, and split it by site. Most of the variance of a held-out score comes from the site (median ICC 0.72–0.95), so extra slides or extra training runs from the same sites do little. A pilot study gives σ_{Δ} , and Eq. (9) then says how many sites a validation needs. In our tumor cohorts, confirming a gain of 0.05 AUROC needed 7 to 53 sites.

Step 3: train once, with a strong recipe Start from a pathology foundation model, fine-tune it end to end with H&E color jitter, and tune the learning rate for that initialization. In our data, color jitter improved mean AUROC over ERM in 18 of 20 held-out domains at the standard tier, and the fine-tuned DINO ViT with jitter had the best worst-site AUROC averaged over the four cohorts (0.895). Two warnings apply. Frozen foundation-model features were not enough for mitotic figures (the frozen probes ranked 18th to 21st on both mitosis cohorts), so frozen features must be checked on the task at hand. And an initialization with the wrong learning rate can fail outright: Barlow Twins fine-tuning with the ImageNet learning rate reached 0.715 mean AUROC on MIDOG++, against 0.891 with its own rate. Domain-generalization objectives deserve low priority; under the primary selection rule, none of the six reached the top five on any cohort.

Step 4: adapt to the new site without labels, if the pipeline allows it Two techniques need neither labels nor any change to the trained weights. They use only unlabeled images of the new site at prediction time. Re-estimating the batch-normalization statistics (BN-adapt) helped most where the shift was severe. At the standard tier it raised mean AUROC by 0.120 on the canine scanners, at all five of them, and by 0.015 and 0.022 on Camelyon17 and MIDOG 2021, but it did nothing on MIDOG++ (−0.004). Per-slide stain normalization gave smaller gains than color jitter in three of the four cohorts at the standard tier. Both techniques must be implemented with care. The test images must arrive in shuffled, mixed batches, because a batch of one class corrupts the statistics (Section 4.4). Stain must be estimated from enough tissue: a per-tile estimate failed at a whole Camelyon17 center (0.557 against 0.976 for the per-slide estimate; Supplementary Section S9). TENT, by contrast, updates model parameters at test time and is thus a form of fine-tuning.

Step 5: select and validate without touching the target sites Choose hyperparameters and checkpoints on held-out data from the training sites only. Validate by holding out whole sites in turn with at least three seeds per model. Report every site, the worst site, the seed spread, the calibration at each site and confidence intervals that account for the training data shared between folds. At the standard scale, the choice of selection rule barely changed the rankings

Table 7

The roadmap of Fig. 9 with its evidence. Numbers are standard-tier results under ID-validation selection unless stated. Grades: strong, holds in all four cohorts at the standard tier; moderate, consistent direction but not significant after correction, or holding with exceptions; mixed, large gains in some cohorts and none in others.

Step	What to do	Evidence in this study	Grade
1. Define the target	Fix the task, sites, scanners and a worst-site acceptance level before training.	Canine ERM: mean 0.785, worst scanner 0.534; the worst site differed between methods (Sec. 5.1).	strong
2. Collect sites	Split by site; size the number of sites from a pilot estimate of σ_{Δ} .	Median ICC 0.72–0.95; a 0.05 gain needs 7–53 sites (tumor) or 3 (mitosis); extra seeds help little (Sec. 5.2, 5.3).	strong
3. Train once, strongly	Fine-tune a pathology foundation model with H&E jitter; tune its own learning rate; check frozen features on the task; give DG objectives low priority.	Jitter better than ERM in 18/20 domains; DINO FT + jitter best mean worst-site AUROC (0.895); frozen probes 18th–21st on mitosis; BT with inherited learning rate 0.715 vs 0.891 on MIDOG++; no DG objective in the top five (Sec. 5.4).	moderate
4. Adapt without labels	Optionally re-estimate BN statistics on shuffled site images, or normalize stain per slide; never per small tile.	BN-adapt +0.120 on canine (5/5 scanners), +0.015 and +0.022 on Camelyon17 and MIDOG 2021, –0.004 on MIDOG++; per-tile Macenko 0.557 vs 0.976 per slide at one center (Suppl. S9).	mixed
5. Select and validate	Select on training-site data; hold out whole sites with ≥ 3 seeds; report every site, the worst site, seeds, ECE and intervals.	Selection rules agree at the standard tier ($\tau_b \geq 0.83$) but not at the compact tier (down to 0.32); ECE 0.027–0.186 on Camelyon17 (Sec. 5.6).	strong
6. Accept each new site	Label a small random sample; accept if the site acceptance test gives $P(\text{AUROC} \geq T) \geq 0.95$; plan m with its planner; monitor afterwards.	Development results alone: 8.6% false acceptance at $T = 0.80$. Test: false acceptance $\leq 2.1\%$; at $T = 0.80$, 84% power with 100 labels, which local labels alone need 200 for; worst subgroup 8.7% and 5.7% with 25 and 50 labels, 3.3% with 100 (Sec. 5.7). Monitoring not tested.	moderate

($\tau_b \geq 0.83$). With the small compact model it did (τ_b down to 0.32), which is one more reason not to draw conclusions from toy models. Calibration also varies: on Camelyon17 alone, the mean ECE of standard-tier methods ranged from 0.027 to 0.186.

Step 6: accept each new site with a small labeled sample, then monitor Before a model is used at a new site, label a small random sample of that site’s patches, spread over many slides, and run the site acceptance test of Section 3.7 against the target of Step 1. The development results of Step 5 supply its prior, and its planner gives the number of labels to collect. At the standard tier, the test kept false acceptance at or below 2.1% and, for targets of 0.80 and 0.85, needed about half as many labels as local labels alone. The development results alone were not enough: they would have accepted 8.6% of the sites that fell below a target of 0.80. Where the development sites score far above the target, a site that fails is a surprise to the prior, so at least 100 labels should be collected; the robust variant, which we added after seeing this case, lowers the risk further. Where they score near or below the target, the prior adds little and local labels decide. After acceptance, inputs and outputs should be monitored; we did not test monitoring.

What the roadmap does not promise It does not guarantee that a model will work at a new site; it raises the chance, measures the risk and checks each site before use. Its evidence comes from patch classifiers at 96 to 224 px on four public research cohorts, not from whole-slide or detection systems in clinical use, and clinical deployment additionally needs prospective and regulatory validation that this study does not provide. The method choices in Step 3 were the best on average, not on every cohort, which is exactly why Steps 5 and 6 cannot be skipped.

6.3. What method developers should report

For researchers who propose or compare methods, the results add five reporting rules to the roadmap. Each one changed a conclusion in this study.

1. *Report every domain and repeat runs.* A single official split is one sample from a wide distribution, and the seeds of one method spanned up to 0.48 AUROC at a single domain.
2. *Report intervals and power.* State σ_{Δ} and the number of domains a claimed gain would need, treat a large non-significant gain as unresolved, and account for the training data that folds share (Nadeau and Bengio, 2003; Bates et al., 2024).
3. *Include controls and tune every method.* On MIDOG++, compute-matched ERM was the best compact-tier arm and well ahead of SimCLR, which passes the same number of images through the encoder; on the canine cohort it was the worst. Without the control, neither result could be attributed correctly.
4. *Report all model-selection rules, and separate transductive methods.* Methods that use unlabeled test-site images were among the strongest and should be compared with each other.
5. *Do not transfer rankings across tasks, and release per-run records.* Cohorts that shared only the shift type ranked methods almost independently at the standard tier, and stored records let others add rules and metrics without retraining.

6.4. Case study: a single-hospital synergy claim that did not survive

This study began with a claim of our own that the protocol of Section 3 overturns. The numbers below come from our earlier (v1) pipeline: a compact CNN on 64 px crops, five epochs and selection on the OOD-validation hospital, not the v2 protocol.

The original claim The model was trained on Camelyon17 hospitals 0, 3 and 4, selected on hospital 1 and tested on hospital 2, the official WILDS split (Koh et al., 2021). With one seed per arm, test AUROC was 0.931 for ERM, 0.895 for SimCLR, 0.826 for SAM and 0.951 for SimCLR+SAM. The interaction $I = \delta_{\text{SimCLR+SAM}} - \delta_{\text{SimCLR}} - \delta_{\text{SAM}}$, with δ the difference from ERM, was +0.161 ($\delta_{\text{SimCLR}} = -0.037$, $\delta_{\text{SAM}} = -0.105$, $\delta_{\text{SimCLR+SAM}} = +0.019$). Each method alone was worse than ERM and the pair was better, so we read I as a synergy. Patch-bootstrap intervals were narrow (SAM-ERM: [-0.108, -0.102]).

What the design got wrong The patch bootstrap measures test sampling, not run-to-run variation. A second ERM seed in the same experiment scored 0.881, 0.050 below the first and about 15 times the half-width of that interval. With both available seeds for ERM and SimCLR, I falls to +0.112. As a contrast of four single runs, I carries the run noise of all four arms.

How the protocol detects it The v1 grid has all five hospitals and three seeds per cell (fold t tests on hospital t). At hospital 2, I is -0.019. Over hospitals 0–4 it is -0.012, -0.012, -0.019, +0.180 and -0.050: negative at four of five, with mean +0.017 (95% CI [-0.097, +0.132], $p = 0.70$, Eq. 3). The single positive value comes from one SAM run at hospital 3 with test AUROC 0.407 (other seeds: 0.908 and 0.864). At hospital 2, the seed SDs of ERM, SAM, SimCLR and SimCLR+SAM (0.105, 0.065, 0.048 and 0.002) give a single-run SD of I of about 0.13 if run noise is independent across arms, so +0.161 is about 1.2 SDs from zero. Seed replication thus shows that one run cannot separate the effect from noise, and domain completeness that its sign depends on the test hospital. We withdraw the synergy claim. Under the v2 protocol (compact tier, five seeds per arm, ID-validation selection), the interaction on Camelyon17 is -0.003, -0.007, +0.066, +0.007 and +0.057 at hospitals 0–4, with mean +0.024 (95% CI [-0.019, +0.067], $p = 0.20$). On the other cohorts it is -0.007 (canine), -0.013 (MIDOG 2021) and +0.003 (MIDOG++). Under v2, the official test hospital shows the largest positive value (+0.066). A single-hospital study on that split would again report a synergy that the full design does not support. The small negative value on MIDOG 2021 (95% CI [-0.021, -0.005], unadjusted $p = 0.02$, $K = 3$) is the only interval that excludes zero.

Validity of the v1 records The v1 defects documented in this revision (MixStyle active at evaluation, Macenko as one fixed color transform, RotInv identical to ERM, inactive IRM annealing, an empty GroupDRO group on MIDOG 2021 and the canine cohort, TENT on single-class test batches) do not touch these four arms. One deviation does apply: v1 SAM updated batch-norm running statistics on the perturbed pass instead of the unperturbed one, the reverse of the standard implementation, and v2 corrects it. It affects SAM and SimCLR+SAM in both v1 experiments, by an amount we have not measured. The single-hospital experiment also used an earlier script and a different 30,000-patch training sample than the grid, so its values are not interchangeable with fold 2's.

6.5. A single-institution cohort cannot support domain claims (NCT-CRC-HE)

We also ran the four v1 arms on NCT-CRC-HE-100K (Kather et al., 2018), with two seeds each. The task was tumor epithelium versus all other tissue except background, on the central 64×64 px of each tile. Training (20,000 tiles), ID validation (4,000) and the SimCLR pool (10,226 unlabeled tiles) came from the color-normalized release. There were two test conditions: the same release without color normalization (4,000 tiles), and CRC-VAL-HE-7K (Kather et al., 2019), which comes from 50 other patients of the same tissue bank (class-balanced to 2,466 tiles).

$K = 1$ by construction The tiles have no site labels, so no domain can be held out. The cohort is one domain, and model selection had to use ID validation. Neither test condition changes the institution.

Result All arms reach ID-validation AUROC of 0.963–0.972 (means over two seeds). Without color normalization, ERM scores 0.791, SAM 0.726, SimCLR+SAM 0.724 and SimCLR 0.702 (−0.089 relative to ERM). On CRC-VAL-HE-7K the four arms span only 0.017 (0.927–0.944). ERM has the lowest ECE in both test conditions (0.157 against 0.206–0.235, and 0.027 against 0.032–0.037). Note that SimCLR’s pool saw only normalized tissue, so it never saw the un-normalized appearance it was tested on.

The arithmetic of Section 6.4 recurs. Without normalization, both single methods are worse than ERM, and $I = +0.087$. A tile bootstrap of the two-seed ensembles gives $+0.078$ [0.070, 0.085]. That interval reflects tile sampling only, while SimCLR’s two seeds differ by 0.102 in the same condition (0.752 and 0.651).

Why this is a scope statement With $K = 1$ there are no leave-one-domain-out folds and no between-domain variance σ_D or ICC (Eq. 5). Eq. 3 also needs $K \geq 2$. The result shows that, on one tissue source, SimCLR pre-trained on normalized tiles did not help under an unseen stain appearance or on new patients. It says nothing about institutional shift either way. We report it to set the scope of our conclusions. The SAM batch-norm caveat above applies here too.

6.6. Limitations

Four cohorts, one per cell The task-by-shift analysis has one cohort per cell. Task, species, tissue, label definition and training-set size vary together between cells, and the cohort-level decomposition is descriptive. The two mitosis cohorts share MIDOG 2021’s 150 images. More cohorts per cell would be needed to separate task from shift type statistically.

Few domains With $K = 3$ to 7, the ICC and the paired contrasts have wide intervals, and fold dependence can only be handled as a sensitivity analysis. MIDOG 2021 has no OOD-validation domain.

Model scale and input size The standard tier uses ResNet-50 at 96 px and ViT-S/16 at 224 px. Patch classification at this size is not whole-slide inference or mitotic-figure detection, and larger inputs or detection pipelines may rank methods differently. The two tiers already disagree in part (Section 5.4).

Foundation models HuggingFace was not reachable from our compute environment, so recent large foundation models such as UNI, Virchow or Prov-GigaPath (Chen et al., 2024; Vorontsov et al., 2024; Xu et al., 2024) were not evaluated. The Lunit models are trained on TCGA only and are licensed for non-commercial research.

Tuning Hyperparameters were tuned with one seed per fold, and the compact tier uses a fixed learning rate. Some selected values lie at a grid edge (Supplementary Section S8). A larger search could change individual ranks.

Transductive stain normalization Per-slide Macenko normalization estimates each test slide’s stain from that slide’s benchmark patches, which were sampled from annotated regions. The effect of this tile set on the estimate was small (Supplementary Section S9), but a label-independent tissue sample would match practice better.

Secondary cohorts PatchCamelyon overlaps with Camelyon17 in its hospitals, and NCT-CRC-HE has no site labels, so neither tests institutional shift. The case studies use numbers from our earlier pipeline, which differs from the v2 protocol.

Site acceptance test The test assumes that a new site is exchangeable with the development sites, so a site unlike all of them must be caught by the local labels; in the hardest subgroup it needed 100 labels to keep false acceptance below 5%. It was validated on labeled patches drawn at random from each site’s test patches, whereas real labels

come in clusters from a few slides, which widens the local uncertainty. It addresses patch-level AUROC, not a clinical endpoint, and its label planner was slightly optimistic at the standard tier.

Metric AUROC is the primary metric. It ignores the decision threshold, which matters in practice, and threshold-dependent metrics may rank methods differently. They are stored in every record.

7. Conclusion

We asked whether pathology models can be trained once and then used at new hospitals and scanners without retraining, and how a practitioner can tell. Holding out every domain of four public cohorts, repeating every run and selecting models without test data, we found that the train-once strategy can work well: the best standard-scale method of each cohort kept a worst-site AUROC between 0.88 and 0.99 without seeing labels from the test site. It is not guaranteed, however. No method was best on every cohort, ERM fell to near chance on one scanner, and with three to seven sites few differences between methods could be confirmed, because most of the uncertainty comes from the sites, not from the seeds.

Fine-tuned pathology foundation models with color augmentation were the most reliable recipe on average, and color augmentation alone was close behind. Re-estimating batch-normalization statistics on unlabeled images of the new site helped mainly under severe scanner shift. Under the primary selection rule, domain-generalization objectives never reached the top five, and frozen foundation-model features were weak for mitotic figures. At the standard scale, rankings agreed more between cohorts that share a task than between cohorts that share a type of shift, so evidence from one task transfers poorly to another.

Because development results cannot guarantee performance at a new site, we proposed a site acceptance test that combines them with a small labeled sample from that site. Validated on every held-out site, it kept false acceptance at or below 2.1% at standard scale and needed about half as many labels as local labels alone, whereas the development results alone accepted 8.6% of the sites that fell below an AUROC of 0.80.

The roadmap of Section 6.2 turns these results into six steps: define a worst-site target, collect many sites, train once with a strong recipe, adapt without labels where the pipeline allows it, validate by holding out whole sites, and accept each new site with the site acceptance test. The protocol, the diagnostics and all per-run records are released so that other cohorts and methods can be added to the same analysis.

CRedit authorship contribution statement

Ayan Chatterjee: [TODO-AUTHORS: confirm these roles] Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. **Amitava Mukherjee:** [TODO-AUTHORS: confirm these roles] Conceptualization, Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and code availability

All data used in this study are public. Camelyon17-WILDS (Koh et al., 2021; Bandi et al., 2019) and PatchCamelyon (Veeling et al., 2018) are released under CC0 1.0. NCT-CRC-HE-100K and CRC-VAL-HE-7K (Kather et al., 2018, 2019; Zenodo, doi:10.5281/zenodo.1214456) are released under CC BY 4.0. The MIDOG 2021 training set (Aubreville et al., 2023a; Zenodo, doi:10.5281/zenodo.4643381) and the multi-scanner canine cutaneous squamous cell carcinoma cohort (Wilm et al., 2023; Zenodo, doi:10.5281/zenodo.7418555) are released under CC BY 4.0. MIDOG++ (Aubreville et al., 2023b; figshare, doi:10.6084/m9.figshare.c.6615571.v1) is publicly available; the

collection record is licensed CC BY 4.0 and its constituent image and label files CC0 1.0. The pretrained pathology weights of Kang et al. (2023) (ResNet-50 Barlow Twins, SwAV and MoCo v2; ViT-S/16 DINO) were obtained from <https://github.com/lunit-io/benchmark-ssl-pathology> and are used under the Lunit Inc. Public License, which permits non-commercial research use only. The ImageNet-1k weights of the ResNet, ConvNeXt and ViT backbones are the torchvision defaults, and the SWAG ViT-B/16 weights (Singh et al., 2022), also obtained through torchvision, are released under CC BY-NC 4.0 (non-commercial use). Code, the fold-materialization scripts for all cohorts, the per-run records of every job, the variance-decomposition, interval and power code, and the script that regenerates every table and figure from those records are available at <https://github.com/InvenTide-AI/histopath-shift>. The release contains the records of all 4,778 runs and the test-set predictions of every final run, from which the site acceptance test is validated. [TODO-AUTHORS: archive the release on Zenodo and cite the code as software with its DOI.]

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